

Kansas Frontier Matrix Domain and Capability Encyclopedia

All Domains, Features, Actions, Viewing Modes, Knowledge Systems, Functions, and Programmable Possibilities

Field	Value
Version	v0.1 PDF-ready master planning manuscript
Date	2026-05-05
Source boundary	Supplied PDFs in /mnt/data; official external source checks listed in Source Ledger; direct workspace scan performed; no mounted KFM repository found.
Truth posture	CONFIRMED / PROPOSED / UNKNOWN / NEEDS VERIFICATION / DENY / ABSTAIN / ERROR.
Evidence limits	Doctrine and source documents are available. Current repo routes, files, tests, workflows, runtime behavior, dashboards, deployments, branch protections and old conversation transcript bodies are not verified in this session.

Executive determination. KFM should be planned and built as a Kansas-first, map-first, time-aware, evidence-first, governed, auditable and reversible spatial knowledge and publication system. The durable public unit of value is the inspectable claim, while maps, tiles, graphs, AI answers, scenes, dashboards and indexes are downstream carriers. This encyclopedia converts the supplied doctrine and domain reports into an implementation-oriented feature/action/view/API/validator/workflow/backlog atlas. Implementation maturity remains UNKNOWN without a mounted repository; all file paths and endpoints are PROPOSED unless explicitly verified.

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2. Executive Summary

KFM is a governed spatial evidence and publication system: Kansas-first in scope, map-first in use, time-aware in data modeling, evidence-first in truth posture, and auditable/reversible in publication. This document exists to turn the supplied doctrine and domain reports into a single product, architecture and implementation reference: domains, features, actions, viewing modes, objects, APIs, validators, workflows, AI behavior, publication controls, correction paths and future programming possibilities.

Exhaustive in this document means systematic coverage of the design space across all named KFM domains and cross-domain systems. It does not mean that every proposed feature is implemented. CONFIRMED claims are bounded to supplied documents, official external checks and current workspace evidence. PROPOSED items are recommended designs. UNKNOWN items require mounted repo, transcripts, tests, logs, workflows, dashboards or runtime evidence.

Current-session evidence confirms a document-rich environment and no mounted Git checkout. Direct commands against /mnt/data returned fatal Git errors, and no .git directory was found in inspected common roots. Therefore this manuscript does not claim existing route names, schemas, CI workflows, services, branch state or deployment maturity.

Old conversation transcripts were not available as transcript artifacts in this session. A personal-context search surfaced prior KFM document references and high-level summaries, not transcript bodies. Those ideas are treated as lineage/doc evidence only where represented in supplied PDFs; otherwise they remain UNKNOWN or PROPOSED.

What is CONFIRMED: core KFM doctrine, Directory Rules responsibility-root discipline, domain-report lineage for hydrology, soil, habitat, fauna, flora, agriculture, geology, atmosphere, hazards, roads/rail/trade, settlements/infrastructure, archaeology, people/DNA/land, MapLibre UI, governed AI, pipeline loop and build strategy. What is PROPOSED: concrete features, APIs, schemas, endpoints, validators, PRs, file paths, UI components and domain thin slices. What remains UNKNOWN: current implementation depth, actual repo conventions, package manager, schema authority in a live checkout, policy engine, workflow enforcement, release system, dashboard existence and steward assignments.

3. Source Ledger and Evidence Method

The source ledger is a control surface, not a bibliography. It states what each source supports and what it cannot prove. Source codes are used throughout the manuscript. External official sources are limited to standards, APIs, tools, source systems and version-sensitive facts that require current verification.

3.1 Current workspace evidence

Probe	Current-session result	Status	Consequence
pwd	/ before cd; /mnt/data inspected as artifact workspace	CONFIRMED	The workspace contains PDFs, not a mounted repo.
git -C /mnt/data status --short	fatal: not a git repository	CONFIRMED	No branch/dirty-state or repo files can be claimed.
git branch --show-current	fatal: not a git repository	CONFIRMED	Current implementation maturity remains UNKNOWN.
find common roots -name .git	no .git directories returned under inspected roots	CONFIRMED	All paths in this document are PROPOSED unless later verified.
/mnt/data file inventory	40 PDF files	CONFIRMED	Supplied PDF corpus is the evidence base for doctrine/lineage.

3.2 KFM and reference source ledger

Source ID	Source name	Type	Status	Supports	Cannot prove	Domains
SRC-DIR	Directory Rules.pdf	doctrine / directory governance	CONFIRMED supplied PDF	Responsibility-root directory law; no root-level domain folders; docs/contracts/schema s/policy/tests/data split.	Cannot prove current repo topology or files.	all domains
SRC-GREEN	Kansas_Frontier_Matrix_Definitive_Greenfield_Building_Plan.pdf	implementation plan	CONFIRMED doctrine / PROPOSED-to-create	Greenfield architecture, build phases, trust spine, public-safe defaults, hydrology-first proof lane.	Assumes empty start; not current repo proof.	system-wide
SRC-BUILD	kfm_build_companion.pdf	implementation companion	CONFIRMED doctrine / PROPOSED mechanics	Operational mechanics: object semantics, identity, evidence closure, validators, rollback drills.	Cannot prove runtime state.	system-wide
SRC-P24	KFM_Components_Pass_24...Manual.pdf	whole-corpus synthesis	CORPUS-CONFIRMED / UNKNOWN implementation	Inspectable claim, artifactization, object-family convergence, external standards verification posture.	Does not prove mounted repo implementation.	all domains
SRC-PIPE	Kansas_Frontier_Matrix_Pipeline_Living_Implementation_Manual_v0.3.pdf	pipeline manual	CONFIRMED doctrine / PROPOSED loop	Lifecycle law; query-save-validate-compile-review-promote-recompile loop; source ledger and object families.	Paths and runtime are proposed without repo.	pipeline / all domains
SRC-IMPLREF	# Kansas Frontier Matrix Implementation Reference.pdf	implementation reference	LINEAGE / NEEDS VERIFICATION	Public repo summary and connector inventory from earlier environment; schema-home ambiguity and hydrology/ecology maturity signal.	Prior connector claims not reverified in this session.	system-wide
SRC-MAP	KFM_MapLibre_Operating_Architecture...REVISED.pdf	UI/map doctrine	CONFIRMED doctrine / PROPOSED implementation	MapLibre 2D shell, renderer boundary, Evidence Drawer and Focus Mode trust surfaces.	No current repo route/component proof.	UI / all domains
SRC-UIAI	KFM_Whole_UI_Governed_AI_Expansion_Report.pdf	UI + AI plan	LINEAGE / PROPOSED	Whole UI expansion, mock governed API, evidence drawer, Focus Mode, review console.	No mounted repo proof.	UI / AI
SRC-GAI	KFM_Governed_AI_Extended_Pro...Report.pdf	governed AI plan	CONFIRMED source scan / PROPOSED design	Provider-neutral adapter, finite outcomes, citation validation, AIReceipt, source-ledger discipline.	No model/runtime implementation proof.	AI / evidence

Source ID	Source name	Type	Status	Supports	Cannot prove	Domains
SRC-OLL	Ollama & Ubuntu Information.pdf	runtime guide	CONFIRMED doctrine / PROPOSED realization	Local model runtime behind governed API; no direct public model traffic; ANSWER/ABSTAIN/DENY/ERROR envelopes.	Version-sensitive runtime details require recheck.	AI runtime
SRC-HYD	KFM_Hydrology_Extended_Pro...Report.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Hydrology proof lane, HUCs, NHDPlus HR, USGS Water Data, NFHL, hydrographs, watershed dashboards.	No current connector proof.	hydrology
SRC-SOIL	kfm_soil_architecture_extended_pro_pdf_only_report.pdf	domain report	CONFIRMED lineage / PROPOSED plan	SSURGO/gSSURGO/SDA, soil horizons/properties, soil moisture, agriculture/hydrology connections.	No mounted soil implementation proof.	soil
SRC-HAB	kfm_habitat_architecture_pdf_only_blueprint_2026-04-21.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Habitat patches, suitability, connectivity, restoration, evidence support, public-safe products.	No current implementation proof.	habitat
SRC-FAUNA	KFM_Fauna_Architecture_PDF_Only_Report.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Taxonomy, occurrence evidence, ranges, geoprivacy, generalized public tiles, stewardship.	No live source connector proof.	fauna
SRC-FLORA	KFM_Flora_Architecture_PDF_Only_Implementation_Blueprint.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Plant taxonomy, specimens, rare plants, invasive plants, phenology, restoration, public-safe redaction.	No current implementation proof.	flora
SRC-AG	KFM_Agriculture_Domain_Implementation_Dossier_REVISED_2026-04-21.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Crops, fields, soils, irrigation, yields, CDL/NASS source roles, suitability and stress indicators.	No repo files verified.	agriculture
SRC-GEO	KFM_Geology_Natural_Resources_Architecture...Report.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Bedrock/surficial geology, boreholes, resources, cross-sections, public-risk controls.	No current implementation proof.	geology
SRC-AIR	KFM_Atmosphere_Air_PDF_Only_Architecture_Report_2026-04-21.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Air quality, weather, smoke, climate, model-vs-observation distinctions, source freshness.	No current source-rights verification.	atmosphere
SRC-HAZ	kfm_hazards_extended_pro_pdf_only_blueprint.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Hazard events, warnings as context only, NFHL, exposure/resilience, non-emergency posture.	No emergency operations proof.	hazards
SRC-ROAD	KFM_Roads_Rail_Trade_Routes_PDF_Only_Architecture_Plan_2026-04-21.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Modern/historic roads, rail, depots, crossings, network graph, sensitive route generalization.	No repo implementation proof.	roads / rail
SRC-SET	kfm_settlements_infrastructure_extended_pro_plan_2026-04-21.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Settlements, towns, cities, infrastructure assets, service areas, dependencies, public-safe views.	No current implementation proof.	settlements/infrastructure
SRC-ARCH	KFM_Archaeology_Architecture_Plan_PDF_Only.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Archaeology sensitivity, exact-location denial, LiDAR candidates, cultural/steward review, 3D documentation.	No live source/steward permissions verified.	archaeology
SRC-PEOPLE	KFM_People_Genealogy_DNA_Land_Ownership_Architecture_Blueprint.pdf	domain report	CONFIRMED lineage / PROPOSED plan	Assertion-first people/genealogy/DNA/land architecture; DNA and living-person privacy fail closed.	No actual person/DNA records inspected.	people / land
SRC-HF	KFM_Habitat_Fauna_Thin_Slice_Extended_Pro_Blueprint.pdf	thin-slice plan	LINEAGE / PROPOSED	Habitat-fauna proof pattern, one controlled occurrence fixture, EvidenceBundle assignment proof.	No actual published slice confirmed.	habitat / fauna
SRC-PLANET	Planetary_Spatial_Systems_Architecture.pdf	spatial systems manual	CONFIRMED source-bounded / PROPOSED additions	CRS/datum/projection, representation-vs-reality, 2.5D vs 3D, digital-twin boundaries.	Product-specific capabilities need verification.	spatial / 3D

Source ID	Source name	Type	Status	Supports	Cannot prove	Domains
SRC-GISPRIMER	a-primer-of-gis-fundamental...pdf	GIS reference	background reference	Representation/communication, projections, coordinate systems, cartographic choices, analysis concepts.	Does not define KFM doctrine.	spatial foundation
SRC-GISSUCC	GIS Succinctly.pdf	GIS reference	background reference	GIS components, spatial databases, geometry types, OGC concepts, loading/SQL/application basics.	Does not prove KFM implementation.	spatial foundation
SRC-ARCGIS	Earth, Space, and Environmental Science Explorations with ArcGIS Pro ed2.pdf	GIS/environment reference	background reference	Flood hazards, hurricanes, wildfire, vulnerability, water resources, climate, field data, planetary exercises.	Tutorial source; not KFM policy.	hydrology / hazards / climate / planetary
SRC-URBAN	GIS in Sustainable Urban Planning and Management.pdf	urban GIS reference	background reference	Urban planning, resilience, participatory GIS, urban growth, flood risk, infrastructure and quality-of-life analysis.	Global examples; not KFM implementation proof.	settlements / hazards
SRC-3DARCH	Archaeological 3D GIS.pdf	archaeology/3D reference	background reference	3D archaeological data, 3D vs 2.5D, surface/subsurface analysis, visibility, volumes.	Does not determine KFM release permissions.	archaeology / 3D
SRC-DDD	Domain-Driven Design Reference.pdf	software architecture reference	background reference	Bounded contexts, entities, value objects, services, modules, published language, context mapping.	Does not define KFM-specific paths.	architecture
SRC-TIME	developing-time-oriented-database-applications-in-sql.pdf	temporal database reference	background reference	Valid time, transaction time, bitemporal tables, temporal keys and queries.	Does not prove KFM database design.	time / data
SRC-SQL	Advanced-SQL-Concepts.pdf	SQL reference	background reference	Window functions, recursive CTEs, joins, set operations, subqueries, analytics.	General; not KFM proof.	analytics
SRC-API	Designing Great Web APIs.pdf	API reference	background reference	API as contracts, resource lifecycles, HTTP verbs/responses, design-first APIs.	Not KFM-specific authority.	API
SRC-REACT	Building User Interfaces...React Programming.pdf	UI reference	background reference	React components, state, events, forms, routing and state management.	No KFM UI implementation proof.	UI
SRC-FULLREACT	fullstack-react-with-typescript.pdf	UI/testing reference	background reference	React + TypeScript, hooks, testing, state, SSR patterns.	General reference.	UI
SRC-WEBPACK	SurviveJS - Webpack and React.pdf	frontend build reference	background reference	Webpack/React build, testing, linting, component structure.	General reference.	UI/build
SRC-CICD	CICD_with_Docker_Kubernetes_Semaphore.pdf	CI/CD reference	background reference	Containers, Kubernetes deployment, CI/CD, rollback and release practices.	Tool-specific details need current verification.	CI/CD
SRC-AIREF	AI_Concepts_Using_Python.pdf	AI reference	background reference	AI/data lifecycle, ML workflows, model evaluation, Python data libraries.	Does not alter governed AI doctrine.	AI
SRC-SCRAPE	instant-web-scraping-with-java.pdf	source acquisition reference	background reference	Scraper error handling, tests, persistence, crawling constraints.	Does not authorize scraping.	source intake

3.3 External official source ledger

Source ID	Source	Type	Status	Supports	Cannot prove
EXT-OAS	OpenAPI Specification v3.2.0	official standard	EXTERNALLY CHECKED	HTTP API contract description.	Version and adoption require recheck before pinning.
EXT-JSON	JSON Schema Draft 2020-12	official schema standard	EXTERNALLY CHECKED	Machine-checkable JSON schemas.	Does not choose KFM schema home.
EXT-DCAT	W3C DCAT Version 3	official catalog standard	EXTERNALLY CHECKED	Catalog metadata and interoperability.	KFM internal fields still proposed.

Source ID	Source	Type	Status	Supports	Cannot prove
EXT-PROV	W3C PROV-O	official provenance ontology	EXTERNALLY CHECKED	Provenance terms and graph/cat closure.	Requires KFM profile.
EXT-STAC	OGC STAC	official geospatial asset standard	EXTERNALLY CHECKED	Spatiotemporal asset cataloging.	KFM catalog profile needed.
EXT-GEOPARQUET	GeoParquet 1.1.0	official/incubating geospatial storage spec	EXTERNALLY CHECKED	Columnar geospatial artifact format.	Tool support and KFM profile need verification.
EXT-PMTILES	PMTiles v3 specification	tile archive spec	EXTERNALLY CHECKED	Static public-safe tile archive possibility.	Hosting/range support and license need verification.
EXT-MAPLIBRE	MapLibre GL JS documentation	official renderer docs	EXTERNALLY CHECKED	Browser vector-tile renderer and style spec context.	Package pinning remains NEEDS VERIFICATION.
EXT-USGS-WATER	USGS Water Data APIs	official source docs	EXTERNALLY CHECKED	Hydrology observations, sites and time-series access.	Connector behavior, parameters and rate limits need verification.
EXT-NHDHR	USGS NHDPlus High Resolution	official source docs	EXTERNALLY CHECKED	Hydrography/network identity and flow context.	Identity ambiguity must be classified.
EXT-WBD	USGS Watershed Boundary Dataset	official source docs	EXTERNALLY CHECKED	HUC/watershed geography source role.	Version and source date must be recorded per dataset.
EXT-NFHL	FEMA National Flood Hazard Layer	official source docs	EXTERNALLY CHECKED	Regulatory flood hazard context.	Not observed inundation or emergency guidance.
EXT-SSURGO	USDA NRCS SSURGO/gSSURGO/SDA	official source docs	EXTERNALLY CHECKED	Soil survey map units, components and attributes.	Static survey vs observation distinction required.
EXT-NASS	USDA NASS CDL/Quick Stats	official source docs	EXTERNALLY CHECKED	Agriculture land cover and statistics.	Terms, API key and aggregation rules need verification.
EXT-AQS	EPA AQS/AirData	official source docs	EXTERNALLY CHECKED	Regulatory ambient air quality archive.	Operational freshness and parameter handling required.
EXT-AIRNOW	AirNow API	official source docs	EXTERNALLY CHECKED	Near-real-time public air quality context.	Not regulatory record; official docs warn accordingly.
EXT-NWS	NWS API	official source docs	EXTERNALLY CHECKED	Forecast, alert and observation context.	Not a KFM emergency alerting system.
EXT-GBIF	GBIF API	official source docs	EXTERNALLY CHECKED	Biodiversity occurrence and species data.	Licenses, sensitive records and uncertainty vary by record.
EXT-INAT	iNaturalist API	official source docs	EXTERNALLY CHECKED	Community observation source family.	Not authoritative legal status; usage limits and geoprivacy apply.
EXT-FWS	USFWS ECOS Data Services	official source docs	EXTERNALLY CHECKED	Threatened/endangered status and reports.	Public sensitivity and exact-location policy still needed.
EXT-NATSERVE	NatureServe Explorer API	official source docs	EXTERNALLY CHECKED	Conservation status and biodiversity data.	Controlled access and data-provider permissions matter.
EXT-CENSUS	US Census APIs and TIGER/Line	official source docs	EXTERNALLY CHECKED	Demography and geography versions.	Historical harmonization/crosswalks still proposed.
EXT-GNIS	USGS GNIS	official source docs	EXTERNALLY CHECKED	Official geographic names and populated place source family.	Historical identity and alternatives need review.
EXT-3DEP	USGS 3D Elevation Program	official source docs	EXTERNALLY CHECKED	Terrain and 3D elevation source family.	Resolution, source date and vertical datum must be captured.
EXT-3DTILES	OGC 3D Tiles	official standard	EXTERNALLY CHECKED	Conditional 3D Tiles publication where evidence burden justifies 3D.	Visualization is not truth.

3.4 Evidence method

- Rank order: current user request; supplied KFM doctrine and artifacts; current-session workspace evidence; official external sources for version-sensitive facts; general technical references for background craft only.
- Docs are evidence of doctrine, design intent and lineage. They are not proof that a current repo contains matching files unless a mounted checkout, tests or generated outputs are inspected.
- Where implementation-specific facts are missing, this document uses PROPOSED, UNKNOWN or NEEDS VERIFICATION rather than plausible-sounding implementation claims.
- Every public or semi-public claim-bearing feature is designed to resolve EvidenceRef -> EvidenceBundle, then pass policy, review and release checks before exposure.

4. KFM Operating Law

Law	Operating rule	Source basis
Inspectable claim	The durable public unit of value is a claim whose evidence, spatial/temporal scope, source role, policy posture, review state, release state and correction lineage can be inspected.	SRC-P24, SRC-GREEN
Evidence hierarchy	EvidenceBundle outranks generated language; source records and resolved evidence outrank maps, search indexes, summaries and model outputs.	SRC-GAI, SRC-PIPE
Lifecycle law	RAW -> WORK/QUARANTINE -> PROCESSED -> CATALOG/TRIPLET -> PUBLISHED. Promotion is a governed state transition, not a file move.	SRC-PIPE, SRC-GREEN
Trust membrane	Public clients use governed APIs, released artifacts, catalog records, tile services and EvidenceBundle resolution, not canonical internal stores.	SRC-MAP, SRC-GREEN
Publication gate	Release requires validation, source/rights/sensitivity checks, review, release manifest, correction path and rollback target.	SRC-BUILD, SRC-PIPE
Correction and rollback	Corrections, withdrawals, supersession and rollback cards must remain visible and auditable.	SRC-P24, SRC-GREEN
AI boundary	AI is interpretive and subordinate to evidence, policy, review and release state. Outcomes are ANSWER, ABSTAIN, DENY or ERROR.	SRC-GAI, SRC-OLL
Map-renderer boundary	MapLibre or any renderer is downstream of trust. It renders released artifacts and view state; it is never truth, policy, publication or citation authority.	SRC-MAP
Sensitivity and rights posture	Living persons, DNA/genomics, rare species, archaeology, culturally sensitive places, critical infrastructure, private landowner-sensitive data and unclear rights fail closed.	SRC-PEOPLE, SRC-ARCH, SRC-FAUNA, SRC-DIR

Directory basis for proposed paths. Directory Rules states that root folders must carry repo-wide responsibility, while domain names usually belong under responsibility roots such as docs/domains/, schemas/contracts/v1/domains/, policy/domains/, tests/domains/ and data lifecycle homes. This document therefore does not propose root-level domain folders; all example paths are PROPOSED and subject to mounted-repo ADR verification. [SRC-DIR]

5. Master Domain Atlas

Domain	Purpose	Boundary	Key source families	Main object families	Map / viewing products	First credible thin slice
Spatial Foundation, Cartography, Reference Systems	CONFIRMED doctrine / PROPOSED implementation.	Owens cross-domain spatial conventions, base layers, CRS/projection rules, geometry validation and cartographic representation rules. It does not own hydrology, roads, archaeology, parcels, habitats, or any domain truth.	USGS 3DEP and The National Map; Census TIGER for administrative geometry; GNIS place names; state/local GIS layers; authoritative basemaps; imagery/DEM/COG sources; historical maps when rights/review permit.	CoordinateReferenceProfile	Kansas overview, base map selector, CRS/scale inspection, source comparison, uncertainty view, stale data view, split map, swipe map, 3D terrain only when evidence burden justifies it.	One Kansas county HUC/road/soil geometry bundle with CRS metadata, geometry fingerprints, validation report, layer manifest and EvidenceBundle-backed feature inspector.
Hydrology	CONFIRMED doctrine / PROPOSED implementation.	Owens watersheds, HUCs, stream reaches, gauges, water level, flow, water quality, hydrologic time series, groundwater context, irrigation/drought links and flood-context overlays. It must not collapse NFHL regulatory flood zones, observed inundation, forecasts and emergency warnings into one truth class.	USGS Water Data APIs, WBD/HUC, NHDPlus HR/3DHP, FEMA NFHL and MSC, 3DEP terrain, state water offices, water quality programs, groundwater wells, irrigation/drought sources.	Watershed	Watershed/HUC drilldown; streams/reaches; gauges with hydrographs; flow/water-level time slider; water quality; groundwater; irrigation links; drought context; NFHL overlay; upstream/downstream tracing; non-emergency flood-context view.	Kansas HUC12 + one USGS gauge fixture + one NHDPlus identity crosswalk + NFHL contextual overlay + hydrograph panel + EvidenceBundle closure + ABSTAIN on ambiguous reach identity.
Soil	CONFIRMED doctrine / PROPOSED implementation.	Owens SSURGO/gSSURGO/SDA-derived map units, components, horizons, soil properties, hydrologic soil groups, erosion, suitability and soil-moisture observations. It links to agriculture, hydrology and habitat but does not own crops, watersheds or species occurrences.	NRCS SSURGO, gSSURGO, Soil Data Access, Web Soil Survey, NASIS-derived tables, pedon databases, Kansas Mesonet soil moisture, remote sensing and erosion/suitability models.	SoilMapUnit	Map unit map, component drilldown, soil profile view, pedon view, HSG layer, erosion/suitability, soil moisture time series, crop-soil-weather view, soil-hydrology joins.	One county SSURGO fixture with map units, components, horizons, HSG, one soil-moisture station sample, profile view and EvidenceBundle-backed layer manifest.
Habitat	CONFIRMED doctrine / PROPOSED implementation.	Owens habitat patches, habitat classes, suitability surfaces, connectivity, corridors, restoration opportunity and stewardship zones. It does not own species occurrence truth, plant taxonomy or fauna taxonomy, but joins to them through governed relationships.	NLCD/land cover, state ecological inventories, NatureServe/ecological systems, conservation plans, restoration projects, remote sensing vegetation indices, field surveys and steward-reviewed habitat models.	HabitatPatch	Habitat patch map, suitability surface, connectivity/corridor view, restoration opportunity, uncertainty mode, sensitivity-redacted mode, habitat-fauna/flora join view.	One NLCD-derived Kansas habitat patch fixture plus one fauna occurrence association and uncertainty/citation report, public generalized tile only.

Domain	Purpose	Boundary	Key source families	Main object families	Map / viewing products	First credible thin slice
Fauna	CONFIRMED doctrine / PROPOSED implementation.	Owns fauna taxonomy, occurrence evidence, range/seasonal range, nesting/den/roost/spawning sites, migration, mortality, disease/pathogen observations, invasive species and conservation status. It must not expose exact sensitive locations publicly.	KDWP/steward sources, USFWS ECOS, NatureServe, GBIF, eBird, iNaturalist, EDDMapS, disease surveillance, wildlife mortality and survey programs. Source role constraints distinguish legal status, occurrence aggregator, model and observation.	Taxon	Species page map, generalized occurrence density, range and seasonal range, migration, abundance/richness, habitat association, disease/mortality context, steward exact-location view, public redacted view.	One non-sensitive public occurrence fixture joined to a habitat patch, with exact point kept steward-only if needed and public generalized layer proven by redaction receipt.
Flora	CONFIRMED doctrine / PROPOSED implementation.	Owns plant taxa, specimen/occurrence evidence, rare plants, vegetation communities, invasive plants, phenology, range and habitat associations. It links to habitat, agriculture and land stewardship but does not own animal records or crop operations.	Herbaria/specimen portals, GBIF, iNaturalist, NatureServe, USFWS, state rare plant programs, vegetation surveys, remote sensing vegetation indices, restoration project records.	PlantTaxon	Plant species pages, generalized occurrence/range, vegetation community, invasive plant spread, phenology calendar, vegetation index, restoration planting, public-safe rare plant product.	One common species occurrence/specimen fixture and one vegetation community polygon with EvidenceBundle-backed species page and public-safe map.
Agriculture	CONFIRMED doctrine / PROPOSED implementation.	Owns agricultural observations, crop/field/rotation/suitability/economy/supporting indicators. It must not publish private farm operations, field-level sensitive details or source-rights-limited data without review.	USDA NASS CDL and QuickStats, NRCS conservation practice data, SSURGO, irrigation/water use, weather/soil moisture, crop insurance/market/economy sources where permitted, local extension sources.	CropObservation	Crop/soil/weather dashboard, CDL crop map, public-safe county/HUC aggregation, irrigation context, suitability analysis, drought/pest/stress indicators, conservation practices where permitted.	County-level crop-year panel using CDL/QuickStats plus SSURGO suitability and Kansas Mesonet weather fixture, with field-level detail denied by default.
Geology and Natural Resources	CONFIRMED doctrine / PROPOSED implementation.	Owns bedrock, surficial geology, geologic age, structures, geomorphology, boreholes, well logs, cores, geophysics, geochemistry, minerals, oil/gas/resource deposits, extraction and reclamation context. It links to hydrology via hydrostratigraphy.	Kansas Geological Survey, USGS geologic maps, borehole/well log repositories, geophysics, geochemistry, mining/oil/gas records, reclamation programs, 3DEP terrain.	GeologicUnit	Bedrock and surficial geology, lithology/age, structures, boreholes/well logs, resources/extraction, reclamation, hydrostratigraphy, cross-section, 3D subsurface view, uncertainty mode.	One county geologic unit fixture with borehole/cross-section evidence, public-safe generalized resource context and EvidenceBundle-backed unit inspector.
Atmosphere, Air, and Climate	CONFIRMED doctrine / PROPOSED implementation.	Owns station observations, air quality parameters, smoke/AOD context, weather station data, mesonet data, wind/precipitation/temperature, climate normals/anomalies and forecast/model fields. It does not replace official advisories or emergency alerting.	EPA AQS/AirData, AirNow, NOAA/NWS, Kansas Mesonet, OpenAQ-like observations if rights allow, satellite AOD/smoke products, climate normals, model fields and advisories as contextual sources.	AirStation	Air quality stations, PM2.5/ozone, smoke/plume map, wind/precip/temp time slider, climate normals/anomalies, station time series, advisory context with official-source redirection.	One station time series fixture with PM2.5/ozone, unit conversion receipt, freshness badge and non-emergency disclaimer.

Domain	Purpose	Boundary	Key source families	Main object families	Map / viewing products	First credible thin slice
Hazards	CONFIRMED doctrine / PROPOSED implementation.	Owns historical events, severe weather, flood, wildfire, smoke, drought, earthquake, heat/cold, hail/wind/tornado, disaster declarations, warnings/advisories as context only, exposure/resilience summaries and hazard timelines. It must redirect life-safety action to official sources.	NOAA Storm Events, NWS API, FEMA disaster declarations/NFHL, USGS earthquakes, USGS Water, NASA FIRMS/smoke products, drought monitors, state emergency management and resilience plans.	HazardEvent	Hazard event timeline, NFHL flood context, drought map, wildfire/smoke, earthquake, severe weather, heat/cold, exposure analysis, resilience summary, not-for-life-safety official-link mode.	Historical flood/severe weather event fixture plus NFHL context and exposure summary, with warning feeds disabled or contextual-only.
Roads, Rail, and Trade Routes	CONFIRMED doctrine / PROPOSED implementation.	Owns modern roads, historic roads, wagon/military/emigrant/stage/cattle trails, rail corridors, depots, sidings, yards, crossings, bridges, ferries, river crossings, freight corridors and network graph projections. It must not expose culturally sensitive corridors or confuse route narrative with surveyed alignment.	KDOT/state/local road data, rail/operator records, historical maps, archives, GNIS, Census/TIGER, county atlases, military/emigrant trail sources, bridges/ferries/crossings records, steward sources for sensitive cultural corridors.	RoadSegment	Modern road/rail, historic route overlay, route time slider, network graph, depot/crossing/ferry points, source comparison, generalized sensitive route view, movement-story view.	One historic route segment with two source geometries, uncertainty class, generalization transform and movement-story node linked to a settlement.
Settlements, Cities, and Infrastructure	CONFIRMED doctrine / PROPOSED implementation.	Owns municipalities, census places, townsites, ghost towns, forts, depots, missions, reservation communities, infrastructure assets, networks, nodes, segments, facilities, service areas, operators, condition observations and dependencies. It must not expose sensitive infrastructure precision by default.	Census/TIGER, GNIS, state/local GIS, municipal records, historical gazetteers/maps, infrastructure operator records, FEMA context, roads/rail records, fort/mission/reservation community archives.	Settlement	Settlement timeline, cities/places, townsites/ghost towns, infrastructure public-safe view, service areas, dependencies, urban growth, annexation, historical identity comparison.	One settlement timeline with GNIS/Census/historical map evidence, alias handling, one public-safe facility/service area and EvidenceBundle-backed inspector.
Archaeology and Cultural Heritage	CONFIRMED doctrine / PROPOSED implementation.	Owns archaeological sites, surveys, artifacts, features, contexts, excavation units, remote sensing anomalies, LiDAR candidates, geophysics, 3D documentation, collections/accessions and chronology. It must not publish exact sites, sacred places, burial/human remains or culturally sensitive material without proper review.	State historic preservation/steward records, tribal/steward review, excavation reports, survey records, museum/collection accessions, LiDAR/remote sensing/geophysics, historical maps, 3D documentation.	ArchaeologicalSite	Public generalized archaeology view, steward exact-site review, survey coverage, candidate anomalies, 3D site documentation, chronology timeline, uncertainty/candidate-vs-confirmed view.	Synthetic archaeology candidate fixture with exact geometry denied, public generalized tile, steward review record and correction/rollback path.

Domain	Purpose	Boundary	Key source families	Main object families	Map / viewing products	First credible thin slice
People, Genealogy, DNA, and Land Ownership	CONFIRMED doctrine / PROPOSED implementation.	Owns person assertions, identity resolution, genealogy relations, family groups, events, residences, migrations, land ownership assertions, deeds/title instruments, assessor/tax records, parcels, temporal ownership and DNA match evidence. It must not infer living-person/DNA relationships publicly without policy approval, and it must not equate assessor records or parcel geometry with title truth.	Public historical records, census, vital records where public/legal, deeds/title instruments, land patents, assessor/tax records, parcel geometry, genealogical trees as hypotheses, DNA match evidence only under strict restricted policy.	PersonAssertion	Public-safe family/history story map, residence/migration timeline, land assertion timeline, parcel/title distinction inspector, restricted DNA review view, correction and dispute view.	Historical deceased-person family/land story using public records only, with relationship and ownership assertions cited and living-person /DNA surfaces denied.
Frontier Demography, Economy, Settlement, Land, and Time Matrix	CONFIRMED doctrine / PROPOSED implementation.	Owns county-year panels, geography versions, frontier definitions, population/economic/agricultural/access observations, settlement status, public-land/land office data, rail/road/market access, administrative boundary changes, uncertainty classes, crosswalks and bitemporal releases.	Census decennial/ACS/historical datasets, Census/TIGER geography, GNIS, NASS, land office/public land records, rail/road access sources, historical gazetteers/maps, economic statistics and county histories.	FrontierDefinition	Frontier county-year matrix dashboard, time slider, county panel map, threshold model comparison, settlement/access overlay, land office/public land layer, uncertainty view.	One historical county-year panel with geography version, population observation, rail/road access observation, frontier threshold model and EvidenceBundle-backed matrix cell inspector.
Planetary, 3D, Digital Twin, and Synthetic Spatial Systems	CONFIRMED doctrine / PROPOSED implementation.	Owns conditional 3D/terrain/globe/scene representation rules, 2.5D-vs-analytical-3D distinction, digital-twin vocabulary, Cesium/3D Tiles/glTF admission when evidence burden requires it, synthetic derived surfaces and view-state separation. It does not own domain truth.	USGS 3DEP, LiDAR/point clouds, 3D Tiles, glTF, terrain services, 3D archaeology data, remote sensing/imagery, synthetic/model surfaces, planetary datasets where rights/source roles permit.	SceneManifest	3D terrain/scene, geology cross-section, archaeology 3D documentation, infrastructure/digital-twin review, planetary map context, synthetic model comparison, representation-vs-reality badge.	A terrain-backed 3D view for one non-sensitive public-safe area with SceneManifest, representation receipt and EvidenceBundle links for every claim-bearing layer.

6. Cross-Domain Capability Taxonomy

Every capability below is programmable, but none is automatically safe. Capabilities become public-facing only after evidence closure, policy decision, review state and release state are satisfied.

Capability	Programmable behavior	Primary objects	Governance rule	Status
Source intake	Register a source candidate with owner, terms, source role, sensitivity, update cadence and retrieval plan.	SourceDescriptor, SourceIntakeRecord	DENY if rights/source role unknown for public use.	PROPOSED unless source registry exists.
Source registry	Maintain approved/retired/quarantined sources and source authority roles.	SourceAuthorityRegister	Source role cannot be invented by AI.	PROPOSED
Rights and terms checks	Evaluate license, attribution, redistribution, privacy and access constraints.	RightsDecision, PolicyDecision	Unknown rights fail closed.	PROPOSED
RAW capture	Store source-native immutable input and checksum.	RawCaptureReceipt	No public RAW access.	PROPOSED
Quarantine	Hold failed, sensitive, rights-unknown or malformed inputs.	QuarantineRecord	Quarantine is not a publishable staging area.	PROPOSED
Normalization	Transform to canonical or domain-normalized candidate records.	TransformReceipt, DatasetVersion	Record transform and loss.	PROPOSED
Validation	Schema, geometry, temporal, rights, sensitivity and evidence checks.	ValidationReport	Fail closed on high-risk ambiguity.	PROPOSED
EvidenceRef generation	Create stable references from claims to evidence bundles.	EvidenceRef	EvidenceRef must resolve before publication.	PROPOSED
EvidenceBundle resolution	Return source list, excerpts/records, provenance, policy/review/release state.	EvidenceBundle	EvidenceBundle outranks generated text.	PROPOSED
Deterministic identity and hashing	Compute source/content/geometry/spec/run hashes and stable IDs.	spec_hash, content_hash, geometry_hash	Do not conflate run hash with content hash.	PROPOSED
Temporal modeling	Track observed, valid, source, retrieval, release and correction time.	TemporalScope, DatasetVersion	ABSTAIN if material temporal scope missing.	PROPOSED
Spatial joins and overlays	Join domains by geometry/time with uncertainty and scale support.	OverlayReceipt	False precision risk.	PROPOSED
Crosswalks	Map identifiers/geographies across versions.	Crosswalk, AmbiguityClass	ABSTAIN on ambiguous identity.	PROPOSED
Graph/triplet projection	Build source-backed relationship projection from canonical/catalog truth.	Triplet, GraphDelta	Graph is derivative.	PROPOSED
Catalog closure	Ensure every published dataset/layer has source, schema, validation, policy and release metadata.	CatalogRecord, CatalogMatrix	No orphan artifacts.	PROPOSED
Layer manifest generation	Declare public-safe map layers, tiles, fields, styles, evidence hooks and policy badges.	LayerManifest	Renderer cannot invent truth.	PROPOSED
Map tile / PMTiles / GeoParquet / COG / 3D Tiles	Generate downstream artifacts with manifests and checksums.	GeoManifest, MapReleaseManifest	Derived artifacts are rebuildable.	PROPOSED
Policy gate	Decide ALLOW / RESTRICT / DENY / ABSTAIN / ERROR for operations.	PolicyDecision	DENY by default for sensitive classes.	PROPOSED
Review console	Queues for source, rights, sensitivity, correction, promotion and rollback review.	ReviewRecord	Reviewer action must be auditable.	PROPOSED
Release manifest	Bind artifacts, validation, policy, review, checksums and rollback target.	ReleaseManifest	Promotion is state transition.	PROPOSED
Correction notice	Record error, correction, withdrawal or supersession.	CorrectionNotice	Corrections visible on public claims.	PROPOSED
Rollback card	Define reversible release rollback target and drill.	RollbackCard	Rollback untested is not reliable.	PROPOSED
Public/Governed API	Serve released evidence-backed payloads and finite response envelopes.	OpenAPI, DTOs	No RAW/WORK/QUARANTINE/direct model path.	PROPOSED

Capability	Programmable behavior	Primary objects	Governance rule	Status
Evidence Drawer	Feature click -> EvidenceBundle -> support, policy, review, release, correction lineage.	EvidenceDrawerPayload	Missing bundle returns ABSTAIN/ERROR.	PROPOSED
Focus Mode	Evidence-bounded QA with finite outcomes.	RuntimeResponseEnvelope , AIRceipt	No uncited claims; DENY sensitive exposure.	PROPOSED
Search and indexes	Search released catalog/evidence/feature summaries.	SearchIndexManifest	Index is derivative.	PROPOSED
Time slider / compare / export	Temporal UI and public-safe export/reporting.	ViewState, ExportReceipt	Export must preserve evidence and policy.	PROPOSED
Monitoring, receipts, audit, dashboards	Operational observability for builds, validation, releases and corrections.	RunReceipt, AuditLog	Dashboards do not replace proofs.	PROPOSED

7. Domain Chapters

Each domain chapter uses the same A-N structure. Status labels are intentionally visible. All API paths and file paths are PROPOSED unless a later mounted-repo inspection verifies them.

7.1. Spatial Foundation, Cartography, Reference Systems

Source basis: SRC-DIR, SRC-GREEN, SRC-P24, SRC-PLANET, SRC-GISPRIMER, SRC-GISSUCC, EXT-GEOPARQUET, EXT-3DEP.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Provide the shared spatial grammar for KFM: CRS, datum, projection, geometry validity, map generalization, scale, resolution, support, uncertainty, spatial indexing and cartographic representation. It must not become a root-level domain bucket or replace evidence, catalog, policy, or release systems.

Owns cross-domain spatial conventions, base layers, CRS/projection rules, geometry validation and cartographic representation rules. It does not own hydrology, roads, archaeology, parcels, habitats, or any domain truth.

B. Source families and source roles

USGS 3DEP and The National Map; Census TIGER for administrative geometry; GNIS place names; state/local GIS layers; authoritative basemaps; imagery/DEM/COG sources; historical maps when rights/review permit.

C. Canonical object families

CoordinateReferenceProfile; GeographyVersion; ProjectionTransformReceipt; GeometryFingerprint; BaseLayerDescriptor; MapStyleRule; ScaleSupportProfile; UncertaintySurface; GeneralizationTransform; LayerManifest; GeoManifest.

D. Spatial and temporal model

Points, lines, polygons, rasters, COGs, tiles, GeoParquet and PMTiles as derivatives. Store CRS, datum, vertical datum, transformation method, spatial support, scale denominator, resolution, source date, retrieval time and release time.

E. Map layers and viewing modes

Kansas overview, base map selector, CRS/scale inspection, source comparison, uncertainty view, stale data view, split map, swipe map, 3D terrain only when evidence burden justifies it.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

CRS transformation audit; geometry validity; topology checks; spatial joins; overlay; buffering with uncertainty; generalization; scale conflict detection; topology gap/overlap detection; raster/vector alignment.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR:
docs/domains/spatial-foundation-cartography-reference-systems/; schemas/contracts/v1/domains/...; policy/domains/...;
tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Spatial Foundation source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Spatial Foundation Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Spatial Foundation time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Spatial Foundation analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Spatial Foundation locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One Kansas county HUC/road/soil geometry bundle with CRS metadata, geometry fingerprints, validation report, layer manifest and EvidenceBundle-backed feature inspector.

7.2. Hydrology

Source basis: SRC-HYD, SRC-PIPE, SRC-P24, SRC-MAP, EXT-USGS-WATER, EXT-NHDHR, EXT-WBD, EXT-NFHL, EXT-3DEP.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Represent Kansas water systems as evidence-bound, time-aware hydrologic features, observations and regulatory contexts without becoming an emergency flood-warning system.

Owns watersheds, HUCs, stream reaches, gauges, water level, flow, water quality, hydrologic time series, groundwater context, irrigation/drought links and flood-context overlays. It must not collapse NFHL regulatory flood zones, observed inundation, forecasts and emergency warnings into one truth class.

B. Source families and source roles

USGS Water Data APIs, WBD/HUC, NHDPlus HR/3DHP, FEMA NFHL and MSC, 3DEP terrain, state water offices, water quality programs, groundwater wells, irrigation/drought sources.

C. Canonical object families

Watershed; HUCUnit; HydroFeature; ReachIdentity; GaugeSite; FlowObservation; WaterLevelObservation; WaterQualityObservation; GroundwaterWell; AquiferObservation; NFHLZone; Hydrograph; UpstreamTrace; WaterUseLink; DroughtLink; IrrigationLink.

D. Spatial and temporal model

Lines for streams, polygons for HUCs/floodplains/aquifers, points for gauges/wells, rasters for terrain and drought. Track observed_time, valid_time, source_time, retrieval_time, release_time, provisional/final status, parameter code, unit, qualifier and uncertainty.

E. Map layers and viewing modes

Watershed/HUC drilldown; streams/reaches; gauges with hydrographs; flow/water-level time slider; water quality; groundwater; irrigation links; drought context; NFHL overlay; upstream/downstream tracing; non-emergency flood-context view.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

HUC rollups; upstream/downstream trace; hydrograph statistics; baseflow/peak comparison; water-quality exceedance context; drought-water link; irrigation dependency; catchment overlay; source ambiguity classification; NFHL exposure overlay.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.

AI behavior	Rule
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/hydrology/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Hydrology source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Hydrology Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Hydrology time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Hydrology analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Hydrology locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

Kansas HUC12 + one USGS gauge fixture + one NHDPlus identity crosswalk + NFHL contextual overlay + hydrograph panel + EvidenceBundle closure + ABSTAIN on ambiguous reach identity.

7.3. Soil

Source basis: SRC-SOIL, SRC-AG, SRC-HYD, EXT-SSURGO, EXT-3DEP.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Provide evidence-bound soil map units, components, horizons, properties and observed soil conditions while separating static survey data from time-varying observations.

Owns SSURGO/gSSURGO/SDA-derived map units, components, horizons, soil properties, hydrologic soil groups, erosion, suitability and soil-moisture observations. It links to agriculture, hydrology and habitat but does not own crops, watersheds or species occurrences.

B. Source families and source roles

NRCS SSURGO, gSSURGO, Soil Data Access, Web Soil Survey, NASIS-derived tables, pedon databases, Kansas Mesonet soil moisture, remote sensing and erosion/suitability models.

C. Canonical object families

SoilMapUnit; SoilComponent; Horizon; SoilProperty; HydrologicSoilGroup; SoilMoistureObservation; ErosionRisk; SuitabilityRating; Pedon; SoilProfileView; ComponentHorizonJoin; SoilTimeCaveat.

D. Spatial and temporal model

Map unit polygons and gridded gSSURGO rasters; profile/horizon depth intervals; station observations with observed_time; static survey valid/interpretation time; source/retrieval/release/correction time.

E. Map layers and viewing modes

Map unit map, component drilldown, soil profile view, pedon view, HSG layer, erosion/suitability, soil moisture time series, crop-soil-weather view, soil-hydrology joins.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Dominant component summarization; horizon depth queries; available water capacity; drainage class and HSG overlays; soil/crop suitability; erosion risk; soil moisture anomaly; soil-flood/agriculture joins.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/soil/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Soil source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Soil Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Soil time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Soil analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Soil locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One county SSURGO fixture with map units, components, horizons, HSG, one soil-moisture station sample, profile view and EvidenceBundle-backed layer manifest.

7.4. Habitat

Source basis: SRC-HAB, SRC-HF, SRC-FAUNA, SRC-FLORA, EXT-GBIF, EXT-NATSERVE.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Model land cover, ecological systems, habitat quality and connectivity as evidence-backed observations and models, with public-safe derivatives for sensitive ecological contexts.

Owns habitat patches, habitat classes, suitability surfaces, connectivity, corridors, restoration opportunity and stewardship zones. It does not own species occurrence truth, plant taxonomy or fauna taxonomy, but joins to them through governed relationships.

B. Source families and source roles

NLCD/land cover, state ecological inventories, NatureServe/ecological systems, conservation plans, restoration projects, remote sensing vegetation indices, field surveys and steward-reviewed habitat models.

C. Canonical object families

HabitatPatch; LandCoverObservation; EcologicalSystem; HabitatQualityScore; SuitabilityModel; ConnectivityEdge; Corridor; RestorationOpportunity; StewardshipZone; ModelRunReceipt; UncertaintySurface.

D. Spatial and temporal model

Polygons and rasters; patch graph edges; modeled suitability with model version, training/source support, spatial resolution, support, uncertainty and release time. Keep model vs observation labels visible.

E. Map layers and viewing modes

Habitat patch map, suitability surface, connectivity/corridor view, restoration opportunity, uncertainty mode, sensitivity-redacted mode, habitat-fauna/flora join view.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Patch metrics; fragmentation; least-cost path/connectivity; habitat suitability; restoration prioritization; habitat/species association; uncertainty comparison; trend by land-cover version.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/habitat/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Habitat source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Habitat Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Habitat time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Habitat analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Habitat locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One NLCD-derived Kansas habitat patch fixture plus one fauna occurrence association and uncertainty/citation report, public generalized tile only.

7.5. Fauna

Source basis: SRC-FAUNA, SRC-HF, SRC-HAB, EXT-GBIF, EXT-INAT, EXT-FWS, EXT-NATSERVE.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Govern animal taxonomy, occurrences, ranges, seasonal ranges, migration, sensitive sites and conservation indicators with strong geoprivacy and steward controls.

Owns fauna taxonomy, occurrence evidence, range/seasonal range, nesting/den/roost/spawning sites, migration, mortality, disease/pathogen observations, invasive species and conservation status. It must not expose exact sensitive locations publicly.

B. Source families and source roles

KDWP/steward sources, USFWS ECOS, NatureServe, GBIF, eBird, iNaturalist, EDDMapS, disease surveillance, wildlife mortality and survey programs. Source role constraints distinguish legal status, occurrence aggregator, model and observation.

C. Canonical object families

Taxon; OccurrenceEvidence; RangePolygon; SeasonalRange; MigrationRoute; SensitiveSite; NestDenRoostSpawningSite; MortalityObservation; DiseaseObservation; InvasiveSpeciesRecord; ConservationStatus; AbundanceIndicator; RichnessIndicator; RedactionReceipt.

D. Spatial and temporal model

Points for occurrences, generalized public tiles for sensitive records, polygons for ranges, lines for migration. Track observation date, event date, source date, retrieval, release and correction time; uncertainty radius and geoprivacy transform.

E. Map layers and viewing modes

Species page map, generalized occurrence density, range and seasonal range, migration, abundance/richness, habitat association, disease/mortality context, steward exact-location view, public redacted view.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Species richness; occupancy/abundance indicators; seasonal range overlap; habitat association; corridor/migration conflict; invasive spread; disease cluster context; geoprivacy transform audit.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/fauna/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Fauna source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Fauna Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Fauna time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Fauna analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Fauna locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One non-sensitive public occurrence fixture joined to a habitat patch, with exact point kept steward-only if needed and public generalized layer proven by redaction receipt.

7.6. Flora

Source basis: SRC-FLORA, SRC-HAB, EXT-GBIF, EXT-INAT, EXT-NATSERVE, EXT-FWS.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Govern plant taxonomy, specimens, occurrences, communities, rare plants, invasives, phenology and restoration planting as evidence-backed botanical claims.

Owens plant taxa, specimen/occurrence evidence, rare plants, vegetation communities, invasive plants, phenology, range and habitat associations. It links to habitat, agriculture and land stewardship but does not own animal records or crop operations.

B. Source families and source roles

Herbaria/specimen portals, GBIF, iNaturalist, NatureServe, USFWS, state rare plant programs, vegetation surveys, remote sensing vegetation indices, restoration project records.

C. Canonical object families

PlantTaxon; SpecimenRecord; FloraOccurrence; RarePlantRecord; VegetationCommunity; InvasivePlantRecord; PhenologyObservation; RangePolygon; HabitatAssociation; BotanicalSurvey; RestorationPlanting; RedactionReceipt.

D. Spatial and temporal model

Specimen/occurrence points with uncertainty and geoprivacy; community polygons; vegetation index rasters; phenology time series; exact rare plant locations fail closed.

E. Map layers and viewing modes

Plant species pages, generalized occurrence/range, vegetation community, invasive plant spread, phenology calendar, vegetation index, restoration planting, public-safe rare plant product.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Phenology trends; plant community classification; invasive spread; restoration suitability; rare plant habitat association; botanical survey completeness; redaction and uncertainty scoring.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.

AI behavior	Rule
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/flora/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Flora source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Flora Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Flora time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Flora analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Flora locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One common species occurrence/specimen fixture and one vegetation community polygon with EvidenceBundle-backed species page and public-safe map.

7.7. Agriculture

Source basis: SRC-AG, SRC-SOIL, SRC-HYD, EXT-NASS, EXT-SSURGO, EXT-3DEP.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Represent crops, fields, soils, irrigation, yields, conservation practices and agricultural economy in public-safe aggregate or permissioned form.

Owns agricultural observations, crop/field/rotation/suitability/economy/supporting indicators. It must not publish private farm operations, field-level sensitive details or source-rights-limited data without review.

B. Source families and source roles

USDA NASS CDL and QuickStats, NRCS conservation practice data, SSURGO, irrigation/water use, weather/soil moisture, crop insurance/market/economy sources where permitted, local extension sources.

C. Canonical object families

CropObservation; FieldCandidate; CropRotation; YieldObservation; IrrigationLink; ConservationPractice; SoilCropSuitability; AgriculturalEconomyObservation; SupplyChainNode; DroughtStressIndicator; PestStressIndicator; AggregationReceipt.

D. Spatial and temporal model

Field polygons may be sensitive; public products aggregate to county/HUC/grid thresholds. Track crop year, growing season, observed/source/retrieval/release time and source-rights constraints.

E. Map layers and viewing modes

Crop/soil/weather dashboard, CDL crop map, public-safe county/HUC aggregation, irrigation context, suitability analysis, drought/pest/stress indicators, conservation practices where permitted.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Crop rotation detection; suitability; yield anomaly; drought stress; irrigation-water dependency; soil/crop/hydrology joins; economic trend; supply chain context; public-safe aggregation tests.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.

AI behavior	Rule
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/agriculture/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Agriculture source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Agriculture Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Agriculture time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Agriculture analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Agriculture locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

County-level crop-year panel using CDL/QuickStats plus SSURGO suitability and Kansas Mesonet weather fixture, with field-level detail denied by default.

7.8. Geology and Natural Resources

Source basis: SRC-GEO, SRC-PLANET, EXT-3DEP, EXT-GEOPARQUET, EXT-3DTILES.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Govern geologic maps, stratigraphy, lithology, structures, subsurface observations and resource contexts without turning interpretations or extraction records into unreviewed truth.

Owns bedrock, surficial geology, geologic age, structures, geomorphology, boreholes, well logs, cores, geophysics, geochemistry, minerals, oil/gas/resource deposits, extraction and reclamation context. It links to hydrology via hydrostratigraphy.

B. Source families and source roles

Kansas Geological Survey, USGS geologic maps, borehole/well log repositories, geophysics, geochemistry, mining/oil/gas records, reclamation programs, 3DEP terrain.

C. Canonical object families

GeologicUnit; Lithology; StratigraphicInterval; GeologicAge; FaultStructure; Borehole; WellLog; CoreSample; GeophysicalObservation; GeochemistrySample; MineralOccurrence; ResourceDeposit; ExtractionSite; ReclamationRecord; CrossSection; HydrostratigraphicUnit.

D. Spatial and temporal model

Polygons for units, lines for structures/cross-sections, points for boreholes/samples, rasters/volumes for geophysics, 3D surfaces/voxels only when justified. Track interpretation version and uncertainty.

E. Map layers and viewing modes

Bedrock and surficial geology, lithology/age, structures, boreholes/well logs, resources/extraction, reclamation, hydrostratigraphy, cross-section, 3D subsurface view, uncertainty mode.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Unit overlay; cross-section generation; borehole correlation; hydrostratigraphy link; resource proximity; geochemistry anomaly; terrain/geomorphology; reclamation status; uncertainty propagation.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/geology-and-natural-resources/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Geology and Natural Resources source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Geology and Natural Resources Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Geology and Natural Resources time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Geology and Natural Resources analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Geology and Natural Resources locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.

Risk	Mitigation
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One county geologic unit fixture with borehole/cross-section evidence, public-safe generalized resource context and EvidenceBundle-backed unit inspector.

7.9. Atmosphere, Air, and Climate

Source basis: SRC-AIR, SRC-HAZ, EXT-AQS, EXT-AIRNOW, EXT-NWS, EXT-3DEP.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Provide air quality, weather, smoke, climate and model context as evidence-labeled observations, official contexts or derived products, not emergency instructions.

Owns station observations, air quality parameters, smoke/AOD context, weather station data, mesonet data, wind/precipitation/temperature, climate normals/anomalies and forecast/model fields. It does not replace official advisories or emergency alerting.

B. Source families and source roles

EPA AQS/AirData, AirNow, NOAA/NWS, Kansas Mesonet, OpenAQ-like observations if rights allow, satellite AOD/smoke products, climate normals, model fields and advisories as contextual sources.

C. Canonical object families

AirStation; AirObservation; PM25Observation; OzoneObservation; SmokeContext; AODRaster; WeatherStation; WeatherObservation; WindField; PrecipitationObservation; TemperatureObservation; ClimateNormal; ClimateAnomaly; ForecastContext; AdvisoryContext.

D. Spatial and temporal model

Station points, raster/model grids, plumes/smoke polygons, time series. Track source freshness, issue/expiry time, observed time, valid time, model run time, units and conversion receipts.

E. Map layers and viewing modes

Air quality stations, PM2.5/ozone, smoke/plume map, wind/precip/temp time slider, climate normals/anomalies, station time series, advisory context with official-source redirection.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Unit conversion; AQI/parameter context; smoke exposure overlay; wind plume context; station anomaly; climate normal departure; drought/hazard/agriculture joins; stale data detection.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/atmosphere-air-and-climate/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Atmosphere source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Atmosphere Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Atmosphere time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Atmosphere analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Atmosphere locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One station time series fixture with PM2.5/ozone, unit conversion receipt, freshness badge and non-emergency disclaimer.

7.10. Hazards

Source basis: SRC-HAZ, SRC-HYD, SRC-AIR, SRC-ARCGIS, EXT-NWS, EXT-NFHL, EXT-USGS-WATER.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Govern historical, regulatory, modeled and operational-context hazard information for analysis and resilience while refusing to act as a life-safety alerting system.

Owns historical events, severe weather, flood, wildfire, smoke, drought, earthquake, heat/cold, hail/wind/tornado, disaster declarations, warnings/advisories as context only, exposure/resilience summaries and hazard timelines. It must redirect life-safety action to official sources.

B. Source families and source roles

NOAA Storm Events, NWS API, FEMA disaster declarations/NFHL, USGS earthquakes, USGS Water, NASA FIRMS/smoke products, drought monitors, state emergency management and resilience plans.

C. Canonical object families

HazardEvent; HazardObservation; WarningContext; AdvisoryContext; DisasterDeclaration; FloodContext; WildfireDetection; SmokeContext; DroughtIndicator; EarthquakeEvent; HeatColdEvent; ExposureSummary; ResilienceSummary; HazardTimeline; ImpactArea.

D. Spatial and temporal model

Event points/lines/polygons, warning/advisory polygons with issue/expiry, exposure zones, regulatory polygons, model grids. Track event time, valid/issue/expiry, source time, retrieval and release time.

E. Map layers and viewing modes

Hazard event timeline, NFHL flood context, drought map, wildfire/smoke, earthquake, severe weather, heat/cold, exposure analysis, resilience summary, not-for-life-safety official-link mode.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Hazard frequency; exposure overlay; event clustering; declaration-event relation; timeline queries; drought/hydrology/agriculture link; stale warning denial; resilience summaries; official-source redirection.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/hazards/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Hazards source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Hazards Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Hazards time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Hazards analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Hazards locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

Historical flood/severe weather event fixture plus NFHL context and exposure summary, with warning feeds disabled or contextual-only.

7.11. Roads, Rail, and Trade Routes

Source basis: SRC-ROAD, SRC-SET, SRC-ARCH, SRC-URBAN, EXT-CENSUS, EXT-GNIS.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Govern modern and historical movement networks as temporal, evidence-backed route, facility, restriction and story objects.

Owns modern roads, historic roads, wagon/military/emigrant/stage/cattle trails, rail corridors, depots, sidings, yards, crossings, bridges, ferries, river crossings, freight corridors and network graph projections. It must not expose culturally sensitive corridors or confuse route narrative with surveyed alignment.

B. Source families and source roles

KDOT/state/local road data, rail/operator records, historical maps, archives, GNIS, Census/TIGER, county atlases, military/emigrant trail sources, bridges/ferries/crossings records, steward sources for sensitive cultural corridors.

C. Canonical object families

RoadSegment; HistoricRoute; RailSegment; Depot; Siding; Yard; Crossing; Bridge; Ferry; RiverCrossing; FreightCorridor; RouteEvent; OperatorStatus; AccessRestriction; NetworkEdge; MovementStoryNode; GeneralizationReceipt.

D. Spatial and temporal model

Lines for routes/segments, points for facilities/crossings, graph edges for traversal, temporal alignment versions and status/operator histories. Uncertainty and generalization are first-class.

E. Map layers and viewing modes

Modern road/rail, historic route overlay, route time slider, network graph, depot/crossing/ferry points, source comparison, generalized sensitive route view, movement-story view.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Network connectivity; travel-time/proximity; temporal alignment; route-status history; market/settlement access; bridge/crossing dependency; source conflict scoring; route generalization audit.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/roads-rail-and-trade-routes/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Roads source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Roads Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Roads time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Roads analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Roads locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One historic route segment with two source geometries, uncertainty class, generalization transform and movement-story node linked to a settlement.

7.12. Settlements, Cities, and Infrastructure

Source basis: SRC-SET, SRC-ROAD, SRC-URBAN, EXT-CENSUS, EXT-GNIS, EXT-NFHL.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Model settlements and infrastructure as temporal places, facilities, networks and service areas with public-safe views for critical infrastructure.

Owns municipalities, census places, townsites, ghost towns, forts, depots, missions, reservation communities, infrastructure assets, networks, nodes, segments, facilities, service areas, operators, condition observations and dependencies. It must not expose sensitive infrastructure precision by default.

B. Source families and source roles

Census/TIGER, GNIS, state/local GIS, municipal records, historical gazetteers/maps, infrastructure operator records, FEMA context, roads/rail records, fort/mission/reservation community archives.

C. Canonical object families

Settlement; Municipality; CensusPlace; Townsite; GhostTown; Fort; Mission; ReservationCommunity; InfrastructureAsset; NetworkNode; NetworkSegment; Facility; ServiceArea; Operator; ConditionObservation; Dependency; AnnexationEvent; UrbanGrowthObservation.

D. Spatial and temporal model

Points/polygons for places, network segments/nodes, service areas, temporal municipal boundaries, annexations and place-identity aliases. Critical assets have precision/access class.

E. Map layers and viewing modes

Settlement timeline, cities/places, townsites/ghost towns, infrastructure public-safe view, service areas, dependencies, urban growth, annexation, historical identity comparison.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Urban growth; annexation timeline; infrastructure dependency; service area coverage; settlement-road/rail access; hazard exposure; place identity reconciliation; public-safe precision audit.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/settlements-cities-and-infrastructure/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Settlements source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Settlements Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Settlements time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Settlements analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Settlements locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One settlement timeline with GNIS/Census/historical map evidence, alias handling, one public-safe facility/service area and EvidenceBundle-backed inspector.

7.13. Archaeology and Cultural Heritage

Source basis: SRC-ARCH, SRC-3DARCH, SRC-PLANET, EXT-3DTILES, EXT-3DEP.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Preserve archaeological and cultural heritage knowledge through strict sensitivity, cultural/steward review, candidate-vs-confirmed distinction and exact-location denial by default.

Owns archaeological sites, surveys, artifacts, features, contexts, excavation units, remote sensing anomalies, LiDAR candidates, geophysics, 3D documentation, collections/accessions and chronology. It must not publish exact sites, sacred places, burial/human remains or culturally sensitive material without proper review.

B. Source families and source roles

State historic preservation/steward records, tribal/steward review, excavation reports, survey records, museum/collection accessions, LiDAR/remote sensing/geophysics, historical maps, 3D documentation.

C. Canonical object families

ArchaeologicalSite; Survey; Artifact; Feature; Context; ExcavationUnit; RemoteSensingAnomaly; LiDARCandidate; GeophysicsObservation; ThreeDDocumentation; CulturalReview; StewardReview; CollectionAccession; ChronologyAssertion; SensitivityTransform.

D. Spatial and temporal model

Exact points/polygons restricted; public generalized/suppressed products only. 3D site models and excavation units require access controls, provenance and transform receipts. Candidate anomalies are not confirmed sites.

E. Map layers and viewing modes

Public generalized archaeology view, steward exact-site review, survey coverage, candidate anomalies, 3D site documentation, chronology timeline, uncertainty/candidate-vs-confirmed view.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Survey coverage; candidate prioritization; viewshed/visibility; stratigraphic/context relations; 3D measurement; chronology uncertainty; looting-risk/sensitivity evaluation; transform audit.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR: docs/domains/archaeology-and-cultural-heritage/; schemas/contracts/v1/domains/...; policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Archaeology and Cultural Heritage source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Archaeology and Cultural Heritage Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Archaeology and Cultural Heritage time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Archaeology and Cultural Heritage analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Archaeology and Cultural Heritage locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.

Risk	Mitigation
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

Synthetic archaeology candidate fixture with exact geometry denied, public generalized tile, steward review record and correction/rollback path.

7.14. People, Genealogy, DNA, and Land Ownership

Source basis: SRC-PEOPLE, SRC-TIME, SRC-API, SRC-DDD, EXT-CENSUS, EXT-GNIS.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Represent people, genealogy, DNA evidence and land ownership as assertion-first, evidence-bound, privacy-aware claims with strong default denial for living-person and DNA/genomic data.

Owns person assertions, identity resolution, genealogy relations, family groups, events, residences, migrations, land ownership assertions, deeds/title instruments, assessor/tax records, parcels, temporal ownership and DNA match evidence. It must not infer living-person/DNA relationships publicly without policy approval, and it must not equate assessor records or parcel geometry with title truth.

B. Source families and source roles

Public historical records, census, vital records where public/legal, deeds/title instruments, land patents, assessor/tax records, parcel geometry, genealogical trees as hypotheses, DNA match evidence only under strict restricted policy.

C. Canonical object families

PersonAssertion; PersonIdentityCandidate; GenealogyRelationship; FamilyGroup; LifeEvent; ResidenceEvent; MigrationEvent; LandOwnershipAssertion; DeedInstrument; TitleInstrument; AssessorRecord; TaxRecord; ParcelVersion; OwnershipInterval; DNAMatchEvidence; RelationshipHypothesis; ReviewRecord.

D. Spatial and temporal model

Residences/parcels/land claims have valid time and source time; parcels are geometry versions, not title proof. Genealogy relations are assertions with evidence and confidence. DNA data restricted by default.

E. Map layers and viewing modes

Public-safe family/history story map, residence/migration timeline, land assertion timeline, parcel/title distinction inspector, restricted DNA review view, correction and dispute view.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Identity resolution; relationship hypothesis scoring; chain-of-title timeline; migration path; residence/land overlap; parcel-version change; source conflict review; privacy/sensitivity risk scoring.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.

AI behavior	Rule
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR:
docs/domains/people-genealogy-dna-and-land-ownership/; schemas/contracts/v1/domains/...; policy/domains/...;
tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	People source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	People Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	People time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain People analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive People locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

Historical deceased-person family/land story using public records only, with relationship and ownership assertions cited and living-person/DNA surfaces denied.

7.15. Frontier Demography, Economy, Settlement, Land, and Time Matrix

Source basis: SRC-IMPLREF, SRC-P24, SRC-TIME, EXT-CENSUS, EXT-GNIS, EXT-NASS.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Build the KFM matrix as a governed spatiotemporal evidence graph and analytical layer, not a flat spreadsheet.

Owns county-year panels, geography versions, frontier definitions, population/economic/agricultural/access observations, settlement status, public-land/land office data, rail/road/market access, administrative boundary changes, uncertainty classes, crosswalks and bitemporal releases.

B. Source families and source roles

Census decennial/ACS/historical datasets, Census/TIGER geography, GNIS, NASS, land office/public land records, rail/road access sources, historical gazetteers/maps, economic statistics and county histories.

C. Canonical object families

FrontierDefinition; GeographyVersion; CountyYearPanel; PopulationObservation; EconomicObservation; AgricultureObservation; AccessObservation; SettlementStatus; LandOfficeRecord; PublicLandRecord; AdminBoundaryChange; Crosswalk; UncertaintyClass; FrontierThresholdModel; MatrixRelease.

D. Spatial and temporal model

County/boundary versions, bitemporal observations, crosswalks between geographies, county-year panels with valid/source/retrieval/release/correction times.

E. Map layers and viewing modes

Frontier county-year matrix dashboard, time slider, county panel map, threshold model comparison, settlement/access overlay, land office/public land layer, uncertainty view.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Frontier threshold classification; population density; market/rail/road access; settlement status transition; county boundary crosswalk; economic/agricultural trend; uncertainty propagation; bitemporal release comparison.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.

AI behavior	Rule
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR:
docs/domains/frontier-demography-economy-settlement-land-and-time-matrix/; schemas/contracts/v1/domains/...;
policy/domains/...; tests/domains/...; fixtures/domains/...; data/raw/, data/processed/, data/catalog/, data/published/.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Frontier Demography source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Frontier Demography Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Frontier Demography time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Frontier Demography analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Frontier Demography locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

One historical county-year panel with geography version, population observation, rail/road access observation, frontier threshold model and EvidenceBundle-backed matrix cell inspector.

7.16. Planetary, 3D, Digital Twin, and Synthetic Spatial Systems

Source basis: SRC-PLANET, SRC-3DARCH, SRC-MAP, EXT-3DTILES, EXT-3DEP, EXT-GEOPARQUET.

A. Mission and boundary

CONFIRMED doctrine / PROPOSED implementation. Govern 3D, terrain, digital-twin and synthetic spatial representations as downstream evidence carriers, never reality or canonical truth.

Owens conditional 3D/terrain/globe/scene representation rules, 2.5D-vs-analytical-3D distinction, digital-twin vocabulary, Cesium/3D Tiles/glTF admission when evidence burden requires it, synthetic derived surfaces and view-state separation. It does not own domain truth.

B. Source families and source roles

USGS 3DEP, LiDAR/point clouds, 3D Tiles, glTF, terrain services, 3D archaeology data, remote sensing/imagery, synthetic/model surfaces, planetary datasets where rights/source roles permit.

C. Canonical object families

SceneManifest; TerrainModel; ThreeDTileSet; GITFAsset; PointCloud; DigitalTwinView; SyntheticSurface; ViewState; RepresentationReceipt; RealityBoundaryNote; ThreeDAdmissionDecision.

D. Spatial and temporal model

2D remains default. 3D uses explicit CRS/vertical datum, precision, source dates, transformation, simplification and evidence burden. Scene state is not authoritative state.

E. Map layers and viewing modes

3D terrain/scene, geology cross-section, archaeology 3D documentation, infrastructure/digital-twin review, planetary map context, synthetic model comparison, representation-vs-reality badge.

F. User actions

browse; search; filter; click feature; open Evidence Drawer; compare sources; submit correction; request access; export public-safe data; generate report; steward review; promotion decision; rollback release. Domain stewards additionally approve promotion, annotate uncertainty, link related objects, run analysis, resolve source conflicts and initiate rollback drills.

G. Analytical functions

Line-of-sight/visibility; volume; cross-section; 3D measurement; terrain derivative; shadow/sun context; model comparison; synthetic scenario audit; vertical uncertainty.

H. Knowledge systems

Knowledge system: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, DecisionEnvelope, ReleaseManifest, LayerManifest, CorrectionNotice, RollbackCard, ReviewRecord, domain ontology, search index, graph projection. The domain ontology defines permitted object types, relationships, source roles, temporal fields, sensitivity classes and finite outcomes. Search, vector retrieval and graph/triples are derivative indexes built from released or review-authorized evidence, not root truth.

I. AI and Focus Mode possibilities

AI behavior	Rule
Allowed	Evidence-bounded summarization over released EvidenceBundles; citation-backed explanation; evidence comparison; steward drafting; anomaly explanation; schema/validator suggestions.
Required abstention	ABSTAIN when EvidenceBundle is missing, citations cannot be validated, source roles conflict, temporal scope is insufficient or user asks for unsupported inference.

AI behavior	Rule
Required denial	DENY direct RAW/WORK/QUARANTINE access, sensitive-location exposure, restricted personal/DNA inference, emergency alerting replacement, or uncited authoritative claims.
Receipt	Emit AIReceipt and RuntimeResponseEnvelope with outcome ANSWER / ABSTAIN / DENY / ERROR, evidence_refs, policy_decision and citation_validation.

J. API, contract, and schema possibilities

Surface	Proposed endpoint / artifact	DTO / schema	Finite outcomes	Status
Domain feature lookup	GET /api/v1/domains/{domain}/features/{id}	FeatureDTO + EvidenceRef list	ANSWER / ABSTAIN / DENY / ERROR	PROPOSED
Evidence bundle	GET /api/v1/evidence/{evidence_ref}	EvidenceBundle	ANSWER / DENY / ERROR	PROPOSED
Layer manifest	GET /api/v1/layers/{layer_id}/manifest	LayerManifest	ANSWER / DENY / ERROR	PROPOSED
Correction submit	POST /api/v1/corrections	CorrectionNoticeCandidate	ACCEPTED / DENY / ERROR	PROPOSED
Review decision	POST /api/v1/review/{queue}/{id}/decision	ReviewRecord	ALLOW / RESTRICT / DENY / ERROR	PROPOSED

PROPOSED file homes, subject to Directory Rules and mounted-repo ADR:
docs/domains/planetary-3d-digital-twin-and-synthetic-spatial-systems/; schemas/contracts/v1/domains/...; policy/domains/...;
tests/domains/...; fixtures/domains/...; data/raw//, data/processed//, data/catalog//, data/published//.

K. Tests and validators

schema validation; source descriptor validation; rights validation; sensitivity validation; evidence closure; temporal logic; geometry validity; policy deny tests; citation validation; release manifest validation; rollback drill; no-network fixtures; non-regression tests. Domain-specific tests must include public-safe redaction/generalization, source-role mismatch denial, stale-state handling and non-regression for prior lineage where relevant.

L. Feature backlog

Group	Feature	Actor/action	Evidence needed	Risk	Validation path	Status
Build first	Planetary source registry + no-network fixture	steward/developer	SourceDescriptor + synthetic fixture	rights/source-role ambiguity	schema/source/rights validators	PROPOSED
Build first	Planetary Evidence Drawer inspector	public/researcher/steward	EvidenceBundle for one feature	uncited public claim	evidence closure + citation tests	PROPOSED
After proof lane	Planetary time slider and compare mode	researcher/steward	versioned observations/layers	false temporal alignment	temporal logic tests	PROPOSED
Ambitious / research	Cross-domain Planetary analytics and graph queries	researcher/AI assistant	source-backed triples and model receipts	derivative becomes truth	graph projection tests	PROPOSED
DENY by default	Unreviewed exact sensitive Planetary locations or private data	public visitor	policy approval + redaction receipt	privacy/public-safety harm	policy deny tests	DENY

M. Risks and mitigations

Risk	Mitigation
Rights uncertainty	Block public release until source terms and redistribution class are recorded.
Sensitive location exposure	Default redaction/generalization, restricted views and geoprivacy transform receipts.
False precision	Show uncertainty/support, scale and source-role badges; abstain on over-precise claims.
Source authority confusion	Use source-role registry and separate observation/model/regulatory/legal/status contexts.
Model hallucination	Citation validation, finite outcomes and no direct model-to-public path.
Stale data	Freshness badges, retrieval/source/release time and stale-state policy.
Rollback complexity	ReleaseManifest + RollbackCard + rollback drill for every release.

N. Domain thin-slice plan

A terrain-backed 3D view for one non-sensitive public-safe area with SceneManifest, representation receipt and EvidenceBundle links for every claim-bearing layer.

8. Cross-Domain Systems Chapters

A. MapLibre / Map Shell

MapLibre is the disciplined 2D renderer inside a governed shell. It renders released layer artifacts and view state; it is not truth, policy, citation, review, release or AI authority. Required UI behaviors include layer registry, source/rights/review/release badges, stale-state badges, correction badges, time control, split/switch/snap-to-feature controls and public-safe tile services. Public layers load only from LayerManifest/ReleaseManifest-approved artifacts. [SRC-MAP, SRC-DIR, EXT-MAPLIBRE]

B. Evidence Drawer

Feature click -> governed API -> EvidenceBundle -> source list -> claim support -> review state -> policy state -> release state -> correction lineage. The drawer is the trust object that turns a map feature into an inspectable claim. Missing or denied evidence returns ABSTAIN, DENY or ERROR rather than a fluent popup. [SRC-MAP, SRC-P24, SRC-GAI]

C. Focus Mode / Governed AI

Focus Mode is evidence-bounded question answering over released or authorized EvidenceBundles. Required finite outcomes: ANSWER, ABSTAIN, DENY, ERROR. Required controls: policy precheck, EvidenceRef resolution, citation validation, policy postcheck, AIReceipt, RuntimeResponseEnvelope and no direct public model client. [SRC-GAI, SRC-OLL]

Outcome	Meaning	Example
ANSWER	Evidence exists, policy allows, citations validate.	Explain a released HUC12 hydrograph.
ABSTAIN	Evidence insufficient or unresolved.	No EvidenceBundle for a historical route alignment.
DENY	Policy or sensitivity blocks response.	Exact rare species or archaeological site location.
ERROR	System failure, malformed request or service problem.	Evidence resolver unavailable.

D. Knowledge Graph and Triplets

The graph is a projection, not canonical truth. Edges are source-backed, typed, temporal and uncertainty-labeled. Relation types include located_in, observed_at, part_of, drains_to, upstream_of, adjacent_to, derived_from, supports_claim, supersedes, corrects, restricts, operated_by, owned_by_assertion, associated_with_habitat and crosswalks_to. [SRC-P24, SRC-PIPE, SRC-DDD]

E. Catalog, Provenance, and Proof Objects

Core objects: SourceDescriptor, EvidenceRef, EvidenceBundle, DatasetVersion, ValidationReport, RunReceipt, AIReceipt, DecisionEnvelope, ReleaseManifest, CatalogMatrix, LayerManifest, GeoManifest, PromotionDecision, CorrectionNotice, RollbackCard and ReviewRecord. Catalog closure requires every artifact to know its source, schema, validation, policy, review, release and rollback context. [SRC-P24, SRC-PIPE, EXT-STAC, EXT-DCAT, EXT-PROV]

F. Query-Save-Recompile Loop

The governed loop is query -> save -> validate -> compile -> review -> promote -> recompile. It must not autonomously publish or mutate truth. It stores auditable summaries, EvidenceResolutionRecords, CandidateDeltas, receipts, validation reports and decisions. Generated text, vector indexes and graph projections are not root truth. [SRC-PIPE]

G. Review Console and Steward Workflows

Review queues cover source activation, rights, sensitivity, evidence closure, schema validation, policy decisions, correction handling, release approvals, denial reasons, rollback drills and audit views. Steward actions write ReviewRecords and DecisionEnvelopes. [SRC-UIAI, SRC-BUILD]

H. Public Product Surfaces

Public map, public API, downloads, reports, story pages, dashboards, educational modules, embedded maps, static atlases and research notebooks consume released artifacts and public-safe EvidenceBundles. Exports carry citations, release version, policy class and correction path. [SRC-MAP, SRC-GREEN]

I. Restricted and Steward Surfaces

Restricted views serve sensitive taxa, archaeology, infrastructure, people/DNA, source-rights-limited materials and steward-only review. Access is deny-by-default, auditable and role-scoped. Public products use suppression, generalization, aggregation or delayed/staged access. [SRC-FAUNA, SRC-ARCH, SRC-PEOPLE, SRC-DIR]

9. Master Feature Matrix

Domain	ingest	validate	normalize	map/time	evidence/graph/AI	review/publish/correct/rollback/export	risk controls
Spatial Foundation, Cartography, Reference Systems	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Hydrology	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Soil	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Habitat	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Fauna	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Flora	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Agriculture	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Geology and Natural Resources	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Atmosphere, Air, and Climate	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Hazards	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Roads, Rail, and Trade Routes	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Settlements, Cities, and Infrastructure	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Archaeology and Cultural Heritage	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
People, Genealogy, DNA, and Land Ownership	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges

Domain	ingest	validate	normalize	map/time	evidence/graph/AI	review/publish/correct/rollback/export	risk controls
Frontier Demography, Economy, Settlement, Land, and Time Matrix	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges
Planetary, 3D, Digital Twin, and Synthetic Spatial Systems	SourceDescriptor + RAW capture	schema/rights/sensitivity/evidence	canonical candidates + transforms	LayerManifest + time slider	EvidenceDrawer + graph projection + Focus Mode	ReleaseManifest, correction, rollback, public-safe export	deny sensitive/unknown rights; stale/uncertainty badges

10. Master Action Matrix

Actor	Allowed actions	Denied actions	Required evidence	Required gates	Outputs
Public visitor	browse public map; search; inspect Evidence Drawer; view stories; download public-safe exports	direct RAW/WORK/QUARANTINE; exact sensitive locations; uncited AI answers	released EvidenceBundle and public ReleaseManifest	public-safe policy, rights, sensitivity, stale-state	map view, report, citation, export receipt
Researcher	advanced search; compare sources; export public/research datasets; notebooks	restricted data without access approval; source scraping without rights	EvidenceBundle, source descriptors, research license class	access role, export policy, citation validation	notebook/export with citations
Steward	restricted review; annotate uncertainty; approve redaction; handle corrections	publish without release manager/review state	source record, policy decision, review record	sensitivity/rights/steward queue	ReviewRecord, DecisionEnvelope
Domain editor	create candidate object; link evidence; propose schema updates; run validations	promote own unreviewed work	source records, fixtures, validation report	schema/policy/evidence gates	CandidateDelta, ValidationReport
Reviewer	approve/deny source activation, policy result, promotion candidate	edit canonical truth without recorded decision	candidate package and validation evidence	separation of duties, audit	ReviewRecord, PromotionDecision
Policy admin	manage policy gates and role classes; review deny reasons	bypass audit or grant unlimited access	policy register and tests	least privilege, policy tests	PolicyDecision, audit log
Release manager	assemble ReleaseManifest; promote; rollback; withdrawal	release without proof/rollback	proof pack, release candidate, rollback card	release gate, signatures/checksums	ReleaseManifest, RollbackCard
Developer	implement schemas/APIs/validators/tests; no-network fixtures	claim production behavior without tests/logs	issues, ADR, tests, fixtures	CI, review, security	PR, test report, docs
AI assistant	summarize released evidence; draft explanations; suggest validators	uncited claims; direct model endpoint; policy override	EvidenceBundle context	pre/post policy and citation validation	RuntimeResponseEnvelope, AIReceipt
Source connector	fetch source under approved descriptor; emit receipts	fetch unknown-rights data to public path	SourceDescriptor and credentials policy	rights/source cadence/terms	RawCaptureReceipt, RunReceipt

11. Master Viewing Mode Atlas

Viewing mode	Purpose / UI behavior	Primary domains
Kansas overview	Public landing map with released public-safe layers, trust badges and stale/correction indicators.	all domains
Domain layer explorer	Select domain layers from released LayerManifest and view source/review/release state.	all
Time slider	Filter valid/observed/source/release time with visible temporal caveats.	time-aware domains
Source comparison	Split or swipe sources/versions with evidence highlights and conflict badges.	all
Evidence-first feature inspector	Feature click opens Evidence Drawer and claim support.	all
Trust badge map	Display evidence, policy, review, release, stale and correction badges.	all
Stale data view	Highlight outdated source/retrieval/release states.	operational/time series
Uncertainty view	Display uncertainty radius/surface/support/confidence classes.	all
Sensitivity-redacted view	Show generalized/suppressed public artifacts and redaction receipts.	fauna/flora/archaeology/infr astructure/people
Split map / swipe map	Compare geographies, source vintages, model outputs or before/after releases.	all
Watershed view	HUC drilldown, gauges, hydrographs, upstream/downstream trace and NFHL context.	hydrology
Soil profile view	Map unit -> component -> horizon profile and pedon/station observations.	soil
Species range view	Generalized occurrence/range/seasonal range with geoprivacy.	fauna/flora
Habitat suitability view	Patch, suitability, connectivity, corridor and uncertainty layers.	habitat
Crop/soil/weather view	Crop, soil suitability, weather/soil moisture and drought stress dashboards.	agriculture
Geology cross-section	Geologic units, boreholes and cross-section/crosswalk evidence.	geology
Air/smoke time series	Station series, plume/smoke context, wind and freshness badges.	atmosphere
Hazard event timeline	Historical/regulatory/context hazard events with non-emergency disclaimers.	hazards
Road/rail network graph	Temporal network edges, facilities, restrictions and movement stories.	roads/rail
Settlement timeline	Place identity, annexation, townsites, infrastructure and service areas over time.	settlements
Archaeology generalized view	Public generalized/suppressed products and candidate-vs-confirmed labels.	archaeology
People/land assertion timeline	Historical public assertions, parcel/title distinction and correction path.	people/land
Frontier county-year matrix	County-year panels, thresholds, crosswalks and uncertainty.	frontier matrix
3D/terrain/scene view	Conditional 3D with representation receipt and evidence burden.	geology/archaeology/planet ary
Narrative story node view	Evidence-backed story nodes linked to features, time and sources.	education/story
Review console view	Queues for source, policy, sensitivity, release, correction and rollback.	steward
Release proof view	ReleaseManifest, proof pack, catalog closure and rollback target inspection.	release

12. Programming Possibilities Backlog

Phase	Description	Domains	Value	Evidence needed	Data needed	Contracts/schemas	APIs/UI	Risks	Rollback path	Status
Phase 0	Source ledger, control plane, schemas, fixtures, validators	all	Establish trust spine before features	Directory Rules, KFM docs, source registry	synthetic no-network fixtures	SourceDescriptor, EvidenceBundle, DecisionEnvelope, ValidationReport	docs + validators + CI	schema/rights drift	revert PR; no public artifact	PROPOSED
Phase 1	Hydrology proof lane	hydrology	First evidence-rich public-safe slice	USGS/WBD/NFHL descriptors	HUC/gauge/NHDPlus/NFHL fixtures	HydroFeature, GaugeObservation, LayerManifest	Map layer + Evidence Drawer	source ambiguity	release candidate rollback	PROPOSED
Phase 2	Soil and habitat/fauna	soil, habitat, fauna	Exercise static survey + sensitive occurrence controls	SSURGO, habitat, occurrence descriptors	soil map unit + habitat + public occurrence	Soil/Habitat/Fauna schemas	profile/species/habitat views	sensitive leakage	disable sensitive layer	PROPOSED
Phase 3	MapLibre UI expansion and Evidence Drawer	all	Make trust visible in map shell	LayerManifest and EvidenceBundle	released layer fixtures	EvidenceDrawerPayload	map shell + badges + drawer	renderer treated as truth	revert layer registry	PROPOSED
Phase 4	Governed AI Focus Mode	all	Evidence-bounded explanations	released EvidenceBundles	QA fixtures with citations	RuntimeResponseEnvelope, AIReceipt	Focus panel	uncited/hallucinated claims	disable AI adapter	PROPOSED
Phase 5	Domain expansion	agriculture, geology, atmosphere, hazards, roads, settlements, archaeology, people	Broaden atlas while preserving controls	domain source registries	domain fixtures	domain schemas/policies	layer/API/backlog	rights and sensitivity	per-domain rollback	PROPOSED
Phase 6	Graph and cross-domain analytics	all	Enable joins and narrative reasoning	source-backed relations	triplet fixtures	GraphDelta, Crosswalk	graph view + queries	graph becomes truth	rebuild graph from catalog	PROPOSED
Phase 7	3D/digital twin where justified	geology, archaeology, infrastructure, planetary	Use 3D only when evidence burden requires	3D Tiles, 3DEP, 3D docs	terrain/scene fixtures	SceneManifest, RepresentationReceipt	3D view	scene false authority	remove scene release	PROPOSED
Phase 8	Public atlas, dashboards, story products	all	Public value and education	released evidence	story/layer fixtures	StoryManifest, ExportReceipt	stories/dashboards/exports	oversimplification	withdraw story/release	PROPOSED
Phase 9	Advanced simulations and model comparisons	selected domains	Scenario analysis with evidence/model separation	model sources, calibration data	model fixture packs	ModelRunReceipt, DecisionEnvelope	notebooks/model comparison	model treated as truth	quarantine model output	PROPOSED

13. Sensitive / Deny-by-Default Register

Class	Examples	Default outcome	Required controls	Source basis
Living persons	Personal data, residences, identity assertions	DENY public exact/identifying output unless legal basis, consent/review and release state are proven	privacy review; redaction; aggregate; staged access	SRC-PEOPLE
DNA/genomics	DNA matches, genomic inference, living-person relatives	DENY by default; restricted steward/research only with policy approval	separate restricted store; no public AI inference	SRC-PEOPLE
Rare species	Exact taxa occurrence/nest/den/oost/spawning sites	DENY public exact location; generalized public products only	geoprivacy transform receipt; steward review	SRC-FAUNA, SRC-FLORA
Archaeology	Site coordinates, burial/sacred/culturally sensitive materials	DENY exact public location by default	cultural/steward review; suppression/generalization	SRC-ARCH
Sacred/culturally sensitive places	Oral history, cultural routes, sacred sites	DENY until steward review and access class approve	consultation record; sensitivity transform	SRC-ARCH, SRC-ROAD
Critical infrastructure	Exact facilities, dependencies, condition observations	RESTRICT/DENY public precision	public-safe aggregation; role-based access	SRC-SET
Private landowner-sensitive data	Field boundaries, owner identity, operations	DENY exact/public if private or rights unclear	aggregation; permissions; policy review	SRC-AG, SRC-PEOPLE
Exact sensitive locations	Any exact point that increases harm risk	DENY by default	redaction/generalization; audit	SRC-DIR
Emergency warning misuse	Operational warnings, forecasts, hazard instructions	DENY life-safety replacement; contextual-only with official redirection	not-for-life-safety disclaimer; issue/expiry freshness	SRC-HAZ, SRC-AIR
Source-rights-limited records	Licensed, restricted, no-redistribution, uncertain terms	DENY public release until terms resolved	rights register; attribution; no public derivative if barred	SRC-BUILD

14. Implementation Roadmap: Small Reversible PRs

PR	Description	Proposed homes	Risk	Acceptance	Rollback
PR-00 no-network fixture	Create synthetic fixtures for SourceDescriptor, EvidenceBundle, LayerManifest, ReleaseManifest and one hydrology object.	tests/fixtures + schemas	No live source, no public data	Fixture validation passes; no network access	Revert PR
PR-01 schema-home ADR	Resolve contracts vs schemas split in mounted repo; state semantic vs machine authority.	docs/adr	Authority drift	ADR accepted; directory rules complied	Supersede ADR
PR-02 source descriptor	Define and validate SourceDescriptor + SourceAuthorityRegister.	contracts/source + schemas + control_plane	Unknown rights/source role	valid/invalid fixtures; deny unknown rights	Disable descriptor
PR-03 validator	Build schema/evidence/rights/sensitivity/temporal/geometry validators.	tools/validators + tests	False confidence	negative tests fail closed	Revert validator or pin previous
PR-04 EvidenceBundle resolver	Resolve EvidenceRef to EvidenceBundle with finite outcomes.	packages/evidence + apps/governed-api	Unresolved evidence	ABSTAIN on missing bundle	Disable route
PR-05 catalog closure	CatalogMatrix, DatasetVersion, ValidationReport and proof closure stub.	data/catalog + schemas + tests	Orphan artifacts	closure test rejects incomplete bundle	Delete release candidate
PR-06 MapLibre layer manifest	LayerManifest + public-safe layer fixture + badges.	apps/web or map shell + data/published	Renderer bypass	map layer reads only released manifest	Remove layer from registry
PR-07 Evidence Drawer	Feature click -> governed API -> evidence drawer payload.	UI + governed API	Popup without evidence	citation/evidence tests	Disable drawer link
PR-08 Focus Mode mock-adapter	Mock AI adapter over released EvidenceBundles only.	runtime/AI + tests	Uncited AI claims	ANSWER/ABSTAIN/DENY/ERROR tests	Disable adapter
PR-09 promotion dry-run	Validate release candidate without public publication.	release + policy + tests	Auto-publish	no-public-write test	Discard candidate
PR-10 rollback drill	Create and run rollback card against a dry-run release.	release/rollback + docs/runbooks	Irreversible release	rollback drill receipt	Restore prior release manifest

Directory Rules basis: all proposed homes are responsibility roots. Domain files belong under docs/domains/, schemas/contracts/v1/domains/, policy/domains/, tests/domains/, fixtures/domains/ and data lifecycle homes. ui/, web/, jsonschema/, policies/, styles/ and viewer_templates/ remain compatibility/transitional unless a mounted repo and ADR prove otherwise. [SRC-DIR]

15. Validation and Acceptance Plan

QA area	Acceptance rule	Validation method	Status
PDF QA	No cut-off tables, readable fonts, source ledger near front, page numbers, searchable text, major chapters page-broken.	Rendered PNG review, pdfinfo, search test.	CONFIRMED for this generated PDF after render QA.
Evidence QA	Every major claim has source tag or truth label; unsupported implementation claims marked PROPOSED/UNKNOWN.	Source ledger audit.	Required before publication.
Domain coverage QA	All named domains and bounded contexts have chapters and thin slices.	Domain self-check matrix.	Required.
Sensitivity QA	Sensitive domains fail closed and have deny/default register entries.	Policy deny tests and register review.	Required.
Schema/contract QA	Contracts, schemas, policy, fixtures and tests are separated.	Directory/ADR check.	Required in repo PR.
Citation QA	Evidence Drawer, Focus Mode and reports cite EvidenceBundle or abstain.	CitationValidationReport.	Required.
Release/rollback QA	ReleaseManifest, PromotionDecision, CorrectionNotice and RollbackCard present.	Promotion dry-run + rollback drill.	Required.
No-overclaim QA	No repo maturity, route names, deployment state or old transcripts claimed without evidence.	Truth-label review.	Required.

16. Appendices

Appendix A. Glossary

Term	Definition
Inspectable claim	A public or semi-public statement whose evidence, source role, spatial/temporal scope, policy posture, review state, release state and correction lineage are inspectable.
EvidenceRef	Stable reference from a claim/object to the evidence supporting it.
EvidenceBundle	Resolved evidence package with source descriptors, supporting records, policy/review/release state and citations.
DecisionEnvelope	Finite decision record for policy, promotion or runtime behavior.
RuntimeResponseEnvelope	AI/API response with outcome ANSWER/ABSTAIN/DENY/ERROR, evidence refs, citations and policy state.
SourceDescriptor	Machine/human source record with authority role, rights, sensitivity, cadence and access facts.
ReleaseManifest	Bound release object listing artifacts, checksums, validation, policy, review and rollback target.
RollbackCard	Auditable instructions and target for reverting or withdrawing a release.
LayerManifest	Public-safe layer descriptor binding tiles/data, fields, style, evidence hooks, source and release state.
GeoManifest	Geospatial artifact manifest for GeoParquet, COG, PMTiles, 3D Tiles or related products.
Focus Mode	Governed AI question-answering mode over released or authorized EvidenceBundles.
Evidence Drawer	UI trust panel opened from a feature that shows evidence, sources, review, policy, release and correction lineage.

Appendix B. Acronym list

Acronym	Meaning
AI	Artificial intelligence
AQS	Air Quality System
API	Application programming interface
CDL	Cropland Data Layer
COG	Cloud Optimized GeoTIFF
CRS	Coordinate reference system
DCAT	Data Catalog Vocabulary
DEM	Digital elevation model
DNA	Deoxyribonucleic acid / genetic evidence class
DTO	Data transfer object
GBIF	Global Biodiversity Information Facility
GIS	Geographic information system
GNIS	Geographic Names Information System
HUC	Hydrologic Unit Code
NFHL	National Flood Hazard Layer
NHDPlus HR	National Hydrography Dataset Plus High Resolution
OPA	Open Policy Agent
PMTiles	Single-file map tile archive format
PROV-O	W3C Provenance Ontology
SDA	Soil Data Access
SSURGO	Soil Survey Geographic Database
STAC	SpatioTemporal Asset Catalog
WBD	Watershed Boundary Dataset

Appendix C. Domain object index

Domain	Object families
Spatial Foundation, Cartography, Reference Systems	CoordinateReferenceProfile; GeographyVersion; ProjectionTransformReceipt; GeometryFingerprint; BaseLayerDescriptor; MapStyleRule; ScaleSupportProfile; UncertaintySurface; GeneralizationTransform; LayerManifest; GeoManifest
Hydrology	Watershed; HUCUnit; HydroFeature; ReachIdentity; GaugeSite; FlowObservation; WaterLevelObservation; WaterQualityObservation; GroundwaterWell; AquiferObservation; NFHLZone; Hydrograph; UpstreamTrace; WaterUseLink; DroughtLink; IrrigationLink
Soil	SoilMapUnit; SoilComponent; Horizon; SoilProperty; HydrologicSoilGroup; SoilMoistureObservation; ErosionRisk; SuitabilityRating; Pedon; SoilProfileView; ComponentHorizonJoin; SoilTimeCaveat
Habitat	HabitatPatch; LandCoverObservation; EcologicalSystem; HabitatQualityScore; SuitabilityModel; ConnectivityEdge; Corridor; RestorationOpportunity; StewardshipZone; ModelRunReceipt; UncertaintySurface
Fauna	Taxon; OccurrenceEvidence; RangePolygon; SeasonalRange; MigrationRoute; SensitiveSite; NestDenRoostSpawningSite; MortalityObservation; DiseaseObservation; InvasiveSpeciesRecord; ConservationStatus; AbundanceIndicator; RichnessIndicator; RedactionReceipt
Flora	PlantTaxon; SpecimenRecord; FloraOccurrence; RarePlantRecord; VegetationCommunity; InvasivePlantRecord; PhenologyObservation; RangePolygon; HabitatAssociation; BotanicalSurvey; RestorationPlanting; RedactionReceipt
Agriculture	CropObservation; FieldCandidate; CropRotation; YieldObservation; IrrigationLink; ConservationPractice; SoilCropSuitability; AgriculturalEconomyObservation; SupplyChainNode; DroughtStressIndicator; PestStressIndicator; AggregationReceipt
Geology and Natural Resources	GeologicUnit; Lithology; StratigraphicInterval; GeologicAge; FaultStructure; Borehole; WellLog; CoreSample; GeophysicalObservation; GeochemistrySample; MineralOccurrence; ResourceDeposit; ExtractionSite; ReclamationRecord; CrossSection; HydrostratigraphicUnit
Atmosphere, Air, and Climate	AirStation; AirObservation; PM25Observation; OzoneObservation; SmokeContext; AODRaster; WeatherStation; WeatherObservation; WindField; PrecipitationObservation; TemperatureObservation; ClimateNormal; ClimateAnomaly; ForecastContext; AdvisoryContext
Hazards	HazardEvent; HazardObservation; WarningContext; AdvisoryContext; DisasterDeclaration; FloodContext; WildfireDetection; SmokeContext; DroughtIndicator; EarthquakeEvent; HeatColdEvent; ExposureSummary; ResilienceSummary; HazardTimeline; ImpactArea
Roads, Rail, and Trade Routes	RoadSegment; HistoricRoute; RailSegment; Depot; Siding; Yard; Crossing; Bridge; Ferry; RiverCrossing; FreightCorridor; RouteEvent; OperatorStatus; AccessRestriction; NetworkEdge; MovementStoryNode
Settlements, Cities, and Infrastructure	Settlement; Municipality; CensusPlace; Townsite; GhostTown; Fort; Mission; ReservationCommunity; InfrastructureAsset; NetworkNode; NetworkSegment; Facility; ServiceArea; Operator; ConditionObservation; Dependency
Archaeology and Cultural Heritage	ArchaeologicalSite; Survey; Artifact; Feature; Context; ExcavationUnit; RemoteSensingAnomaly; LiDARCandidate; GeophysicsObservation; ThreeDDocumentation; CulturalReview; StewardReview; CollectionAccession; ChronologyAssertion; SensitivityTransform
People, Genealogy, DNA, and Land Ownership	PersonAssertion; PersonIdentityCandidate; GenealogyRelationship; FamilyGroup; LifeEvent; ResidenceEvent; MigrationEvent; LandOwnershipAssertion; DeedInstrument; TitleInstrument; AssessorRecord; TaxRecord; ParcelVersion; OwnershipInterval; DNAMatchEvidence; RelationshipHypothesis
Frontier Demography, Economy, Settlement, Land, and Time Matrix	FrontierDefinition; GeographyVersion; CountyYearPanel; PopulationObservation; EconomicObservation; AgricultureObservation; AccessObservation; SettlementStatus; LandOfficeRecord; PublicLandRecord; AdminBoundaryChange; Crosswalk; UncertaintyClass; FrontierThresholdModel; MatrixRelease
Planetary, 3D, Digital Twin, and Synthetic Spatial Systems	SceneManifest; TerrainModel; ThreeDTileSet; GITFAsset; PointCloud; DigitalTwinView; SyntheticSurface; ViewState; RepresentationReceipt; RealityBoundaryNote; ThreeDAdmissionDecision

Appendix D. Source family index

Domain	Source families
Spatial Foundation, Cartography, Reference Systems	USGS 3DEP and The National Map; Census TIGER for administrative geometry; GNIS place names; state/local GIS layers; authoritative basemaps; imagery/DEM/COG sources; historical maps when rights/review permit.
Hydrology	USGS Water Data APIs, WBD/HUC, NHDPlus HR/3DHP, FEMA NFHL and MSC, 3DEP terrain, state water offices, water quality programs, groundwater wells, irrigation/drought sources.
Soil	NRCS SSURGO, gSSURGO, Soil Data Access, Web Soil Survey, NASIS-derived tables, pedon databases, Kansas Mesonet soil moisture, remote sensing and erosion/suitability models.
Habitat	NLCD/land cover, state ecological inventories, NatureServe/ecological systems, conservation plans, restoration projects, remote sensing vegetation indices, field surveys and steward-reviewed habitat models.

Domain	Source families
Fauna	KDWP/steward sources, USFWS ECOS, NatureServe, GBIF, eBird, iNaturalist, EDDMapS, disease surveillance, wildlife mortality and survey programs. Source role constraints distinguish legal status, occurrence aggregator, model and observation.
Flora	Herbaria/specimen portals, GBIF, iNaturalist, NatureServe, USFWS, state rare plant programs, vegetation surveys, remote sensing vegetation indices, restoration project records.
Agriculture	USDA NASS CDL and QuickStats, NRCS conservation practice data, SSURGO, irrigation/water use, weather/soil moisture, crop insurance/market/economy sources where permitted, local extension sources.
Geology and Natural Resources	Kansas Geological Survey, USGS geologic maps, borehole/well log repositories, geophysics, geochemistry, mining/oil/gas records, reclamation programs, 3DEP terrain.
Atmosphere, Air, and Climate	EPA AQS/AirData, AirNow, NOAA/NWS, Kansas Mesonet, OpenAQ-like observations if rights allow, satellite AOD/smoke products, climate normals, model fields and advisories as contextual sources.
Hazards	NOAA Storm Events, NWS API, FEMA disaster declarations/NFHL, USGS earthquakes, USGS Water, NASA FIRMS/smoke products, drought monitors, state emergency management and resilience plans.
Roads, Rail, and Trade Routes	KDOT/state/local road data, rail/operator records, historical maps, archives, GNIS, Census/TIGER, county atlases, military/emigrant trail sources, bridges/ferries/crossings records, steward sources for sensitive cultural corridors.
Settlements, Cities, and Infrastructure	Census/TIGER, GNIS, state/local GIS, municipal records, historical gazetteers/maps, infrastructure operator records, FEMA context, roads/rail records, fort/mission/reservation community archives.
Archaeology and Cultural Heritage	State historic preservation/steward records, tribal/steward review, excavation reports, survey records, museum/collection accessions, LiDAR/remote sensing/geophysics, historical maps, 3D documentation.
People, Genealogy, DNA, and Land Ownership	Public historical records, census, vital records where public/legal, deeds/title instruments, land patents, assessor/tax records, parcel geometry, genealogical trees as hypotheses, DNA match evidence only under strict restricted policy.
Frontier Demography, Economy, Settlement, Land, and Time Matrix	Census decennial/ACS/historical datasets, Census/TIGER geography, GNIS, NASS, land office/public land records, rail/road access sources, historical gazetteers/maps, economic statistics and county histories.
Planetary, 3D, Digital Twin, and Synthetic Spatial Systems	USGS 3DEP, LiDAR/point clouds, 3D Tiles, glTF, terrain services, 3D archaeology data, remote sensing/imagery, synthetic/model surfaces, planetary datasets where rights/source roles permit.

Appendix E. Feature index

Feature/capability	Behavior	Objects	Governance
Source intake	Register a source candidate with owner, terms, source role, sensitivity, update cadence and retrieval plan.	SourceDescriptor, SourceIntakeRecord	DENY if rights/source role unknown for public use.
Source registry	Maintain approved/retired/quarantined sources and source authority roles.	SourceAuthorityRegister	Source role cannot be invented by AI.
Rights and terms checks	Evaluate license, attribution, redistribution, privacy and access constraints.	RightsDecision, PolicyDecision	Unknown rights fail closed.
RAW capture	Store source-native immutable input and checksum.	RawCaptureReceipt	No public RAW access.
Quarantine	Hold failed, sensitive, rights-unknown or malformed inputs.	QuarantineRecord	Quarantine is not a publishable staging area.
Normalization	Transform to canonical or domain-normalized candidate records.	TransformReceipt, DatasetVersion	Record transform and loss.
Validation	Schema, geometry, temporal, rights, sensitivity and evidence checks.	ValidationReport	Fail closed on high-risk ambiguity.
EvidenceRef generation	Create stable references from claims to evidence bundles.	EvidenceRef	EvidenceRef must resolve before publication.
EvidenceBundle resolution	Return source list, excerpts/records, provenance, policy/review/release state.	EvidenceBundle	EvidenceBundle outranks generated text.
Deterministic identity and hashing	Compute source/content/geometry/spec/run hashes and stable IDs.	spec_hash, content_hash, geometry_hash	Do not conflate run hash with content hash.
Temporal modeling	Track observed, valid, source, retrieval, release and correction time.	TemporalScope, DatasetVersion	ABSTAIN if material temporal scope missing.
Spatial joins and overlays	Join domains by geometry/time with uncertainty and scale support.	OverlayReceipt	False precision risk.
Crosswalks	Map identifiers/geographies across versions.	Crosswalk, AmbiguityClass	ABSTAIN on ambiguous identity.
Graph/triplet projection	Build source-backed relationship projection from canonical/catalog truth.	Triplet, GraphDelta	Graph is derivative.
Catalog closure	Ensure every published dataset/layer has source, schema, validation, policy and release metadata.	CatalogRecord, CatalogMatrix	No orphan artifacts.
Layer manifest generation	Declare public-safe map layers, tiles, fields, styles, evidence hooks and policy badges.	LayerManifest	Renderer cannot invent truth.
Map tile / PMTiles / GeoParquet / COG / 3D Tiles	Generate downstream artifacts with manifests and checksums.	GeoManifest, MapReleaseManifest	Derived artifacts are rebuildable.
Policy gate	Decide ALLOW / RESTRICT / DENY / ABSTAIN / ERROR for operations.	PolicyDecision	DENY by default for sensitive classes.
Review console	Queues for source, rights, sensitivity, correction, promotion and rollback review.	ReviewRecord	Reviewer action must be auditable.
Release manifest	Bind artifacts, validation, policy, review, checksums and rollback target.	ReleaseManifest	Promotion is state transition.
Correction notice	Record error, correction, withdrawal or supersession.	CorrectionNotice	Corrections visible on public claims.
Rollback card	Define reversible release rollback target and drill.	RollbackCard	Rollback untested is not reliable.
Public/Governed API	Serve released evidence-backed payloads and finite response envelopes.	OpenAPI, DTOs	No RAW/WORK/QUARANTINE/direct model path.
Evidence Drawer	Feature click -> EvidenceBundle -> support, policy, review, release, correction lineage.	EvidenceDrawerPayload	Missing bundle returns ABSTAIN/ERROR.
Focus Mode	Evidence-bounded QA with finite outcomes.	RuntimeResponseEnvelope, AIReceipt	No uncited claims; DENY sensitive exposure.
Search and indexes	Search released catalog/evidence/feature summaries.	SearchIndexManifest	Index is derivative.

Feature/capability	Behavior	Objects	Governance
Time slider / compare / export	Temporal UI and public-safe export/reporting.	ViewState, ExportReceipt	Export must preserve evidence and policy.
Monitoring, receipts, audit, dashboards	Operational observability for builds, validation, releases and corrections.	RunReceipt, AuditLog	Dashboards do not replace proofs.

Appendix F. Action index

Action	Governance note
browse	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
search	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
filter	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
inspect evidence	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
click feature	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
open Evidence Drawer	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
ask Focus Mode	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
compare sources	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
review candidate	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
flag issue	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
submit correction	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
request access	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
export public-safe data	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
generate report	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
approve promotion	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
rollback release	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
annotate uncertainty	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
link related objects	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.
run analysis	Allowed only when role, evidence, policy, review and release state permit; otherwise ABSTAIN, DENY or ERROR.

Appendix G. Viewing mode index

Viewing mode	Purpose	Domains
Kansas overview	Public landing map with released public-safe layers, trust badges and stale/correction indicators.	all domains
Domain layer explorer	Select domain layers from released LayerManifest and view source/review/release state.	all
Time slider	Filter valid/observed/source/release time with visible temporal caveats.	time-aware domains
Source comparison	Split or swipe sources/versions with evidence highlights and conflict badges.	all
Evidence-first feature inspector	Feature click opens Evidence Drawer and claim support.	all
Trust badge map	Display evidence, policy, review, release, stale and correction badges.	all
Stale data view	Highlight outdated source/retrieval/release states.	operational/time series
Uncertainty view	Display uncertainty radius/surface/support/confidence classes.	all

Viewing mode	Purpose	Domains
Sensitivity-redacted view	Show generalized/suppressed public artifacts and redaction receipts.	fauna/flora/archaeology/infrastructure/people
Split map / swipe map	Compare geographies, source vintages, model outputs or before/after releases.	all
Watershed view	HUC drilldown, gauges, hydrographs, upstream/downstream trace and NFHL context.	hydrology
Soil profile view	Map unit -> component -> horizon profile and pedon/station observations.	soil
Species range view	Generalized occurrence/range/seasonal range with geoprivacy.	fauna/flora
Habitat suitability view	Patch, suitability, connectivity, corridor and uncertainty layers.	habitat
Crop/soil/weather view	Crop, soil suitability, weather/soil moisture and drought stress dashboards.	agriculture
Geology cross-section	Geologic units, boreholes and cross-section/crosswalk evidence.	geology
Air/smoke time series	Station series, plume/smoke context, wind and freshness badges.	atmosphere
Hazard event timeline	Historical/regulatory/context hazard events with non-emergency disclaimers.	hazards
Road/rail network graph	Temporal network edges, facilities, restrictions and movement stories.	roads/rail
Settlement timeline	Place identity, annexation, townsites, infrastructure and service areas over time.	settlements
Archaeology generalized view	Public generalized/suppressed products and candidate-vs-confirmed labels.	archaeology
People/land assertion timeline	Historical public assertions, parcel/title distinction and correction path.	people/land
Frontier county-year matrix	County-year panels, thresholds, crosswalks and uncertainty.	frontier matrix
3D/terrain/scene view	Conditional 3D with representation receipt and evidence burden.	geology/archaeology/planetary
Narrative story node view	Evidence-backed story nodes linked to features, time and sources.	education/story
Review console view	Queues for source, policy, sensitivity, release, correction and rollback.	steward
Release proof view	ReleaseManifest, proof pack, catalog closure and rollback target inspection.	release

Appendix H. AI prompt/action index

AI action	Rule	Outcome
Summarize evidence	Allowed over resolved EvidenceBundles with citations	ANSWER or ABSTAIN
Compare two sources	Allowed when both sources are admissible and source roles visible	ANSWER / ABSTAIN
Explain uncertainty	Allowed with uncertainty/support metadata	ANSWER
Draft correction notice	Allowed as steward aid; human review required	ANSWER / ABSTAIN
Suggest schema/validator	Allowed as developer aid; tests required	ANSWER
Expose exact sensitive location	Denied for rare species/archaeology/infrastructure/private/living-person data	DENY
Make uncited claim	Forbidden	ABSTAIN
Emergency instructions	KFM is not an emergency alerting replacement	DENY
Infer DNA/living-person relationship	Denied unless strict policy approval; public path denied	DENY

Appendix I. Policy gate index

Gate	Purpose
source_role_gate	Rejects unknown or inappropriate source role for claim type.
rights_gate	Rejects public release when license/terms/redistribution are unclear.
sensitivity_gate	Restricts or denies sensitive locations, people/DNA, archaeology, infrastructure and private data.
evidence_closure_gate	Requires EvidenceRef -> EvidenceBundle resolution before claim-bearing release.
temporal_gate	Requires material time fields and handles stale/provisional status.
geometry_gate	Checks CRS, validity, precision, uncertainty, support and generalization.
citation_gate	Rejects generated or public claims without validated citations.
review_gate	Requires steward/reviewer decision for promotion and sensitive releases.
release_gate	Requires ReleaseManifest, proof, correction path and rollback target.
rollback_gate	Requires tested rollback card and release lineage.

Appendix J. Open questions

Question	Status / disposition
Repo topology	UNKNOWN until a real checkout is mounted and scanned.
Schema authority	NEEDS VERIFICATION: contracts/ vs schemas/ split must be confirmed or decided by ADR.
Package manager and frameworks	UNKNOWN: API/UI/runtime stack cannot be claimed from supplied PDFs alone.
Source rights	NEEDS VERIFICATION per source before connectors or public release.
Steward roles	NEEDS VERIFICATION: domain reviewers, policy admins and release managers must be named in repo governance.
Old conversation evidence	UNKNOWN: no transcript artifacts available in this session.
Sensitive data operations	NEEDS VERIFICATION: access controls, audit logs and redaction transforms must be implemented and tested.
Release signatures/proofs	NEEDS VERIFICATION: signing/proof toolchain and workflow enforcement.
Public API paths	PROPOSED only; route names must adapt to repo conventions.

Appendix K. Verification backlog

Backlog item	Verification action	Purpose
Run mounted repo scan	git status, branch, file inventory, package markers, CI/workflow scan	CONFIRM implementation baseline
ADR schema-home	Inspect docs/adr and existing schemas/contracts	Decide authority split
Run no-network fixtures	Validate SourceDescriptor, EvidenceBundle, ReleaseManifest	Establish trust spine
External source rights review	Per source descriptor license/terms/cadence/API behavior	Prevent rights violations
Sensitivity policy tests	Rare species, archaeology, people/DNA, critical infrastructure	Fail closed
Map shell proof	LayerManifest -> map -> feature click -> EvidenceDrawer	Renderer downstream of trust
Focus mock proof	EvidenceBundle -> AIReceipt -> citation validation	AI subordinate to evidence
Rollback drill	Release dry-run -> rollback card -> restored prior manifest	Reversibility

Appendix L. Supersession and lineage notes

This encyclopedia is a synthesis and planning artifact. It supersedes no source doctrine, source report or official standard. It consolidates the supplied KFM domain reports, UI/AI manuals, pipeline manual, Directory Rules, greenfield plan and build companion into an encyclopedia view. Prior scaffold reports and older implementation references are retained as lineage and design pressure; none are upgraded to current implementation proof without a mounted repo, tests, logs or generated artifacts.

Final self-check

Self-check question	Result
1. Are all domains covered?	YES - all named domains and bounded contexts have atlas entries and chapters.
2. Are all cross-domain systems covered?	YES - MapLibre, Evidence Drawer, Focus Mode, graph, catalog/proof, loop, review, public/restricted surfaces are covered.
3. Are features, actions, views, knowledge systems, functions and programming possibilities covered?	YES - capability taxonomy, action matrix, viewing atlas, feature matrix and backlog are included.
4. Are sensitive domains fail-closed?	YES - deny-by-default register and domain-specific denial rules included.
5. Are AI and maps subordinate to evidence?	YES - operating law, MapLibre chapter and Focus Mode chapter make this explicit.
6. Are old conversations used only when actually available?	YES - no transcript artifacts were available; ideas are UNKNOWN/PROPOSED unless present in supplied PDFs.
7. Are unsupported claims labeled?	YES - repo implementation, routes, paths and live maturity are PROPOSED/UNKNOWN/NEEDS VERIFICATION.
8. Are tables readable?	YES - tables are narrow, wrapped and split across pages; major matrices use compact readable rows.
9. Are publication, correction and rollback included?	YES - operating law, source objects, roadmap, matrices and validation plan include them.
10. Is the document useful for future implementation?	YES - it provides source ledger, domain chapters, capability taxonomy, PR roadmap, validators, APIs, thin slices and verification backlog.